

A STRONG DOMINABILITY CRITERION AND AN OKA UNION THEOREM

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ABSTRACT. We prove that an n -dimensional complex manifold Y is Oka if and only if it is strongly dominable: for every $y \in Y$, there is an entire map $F : \mathbb{C}^n \rightarrow Y$ such that $F(0) = y$ and dF_0 is invertible. The argument converts this pointwise domination into the convex approximation property in every source dimension. We also show that the Oka property extends across proper closed complex analytic subsets: if A is such a subset of a connected complex manifold X and $X \setminus A$ is Oka, then X is Oka. The criterion has broad applications and produces many new examples of Oka manifolds.

1. INTRODUCTION

The Oka principle seeks geometric conditions on a complex manifold Y under which continuous maps from Stein spaces to Y may be deformed to holomorphic maps, with approximation and interpolation. Its classical origins lie in Oka's 1939 solution of the second Cousin problem [28] and in Grauert's Oka principle for holomorphic bundles [22]; see the historical account in [16, Section 1]. The modern theory of Oka manifolds began with Gromov's geometric interpretation of holomorphic flexibility. Gromov [23] introduced elliptic manifolds and explained this flexibility through dominating holomorphic sprays. Forstnerič [11, 12, 13] subsequently proved that the approximation, interpolation, and parametric forms of the Oka property are equivalent to convex approximation. This characterization leads to the standard definition [15, Definition 5.4.1]; see [15, 16] for the general theory and its development.

Definition 1.1 ([13]). A complex manifold Y is an *Oka manifold* if it has the *convex approximation property* CAP: for every $m \geq 1$, compact convex set $K \subset \mathbb{C}^m$, holomorphic map $f : U \rightarrow Y$ on an open neighborhood U of K , and $\epsilon > 0$, there is $F \in \mathcal{O}(\mathbb{C}^m, Y)$ such that

$$\sup_{z \in K} d_Y(F(z), f(z)) < \epsilon,$$

where d_Y is any distance inducing the topology of Y .

Thus the Oka property may be tested on compact convex subsets of Euclidean spaces, although its consequences concern maps from arbitrary Stein sources. Unless explicitly stated otherwise, every complex manifold in this paper is connected.

Strong dominability is a pointwise condition: at each point it asks for a globally defined entire map that is locally biholomorphic at a preimage of that point. The Oka property, by contrast, concerns simultaneous approximation for maps from Stein sources of every dimension. Whether the former already forces the latter is a central problem in Oka theory, emphasized by Forstnerič [16, Section 3.3]; it is the main question addressed here.

Definition 1.2. An n -dimensional complex manifold Y is *dominable at* $y \in Y$ if there is a holomorphic map $H : \mathbb{C}^n \rightarrow Y$ such that $H(0) = y$ and the differential $dH_0 : \mathbb{C}^n \rightarrow T_y Y$ is an isomorphism. The manifold Y is *strongly dominable* if it is dominable at every point.

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Every Oka manifold is strongly dominable, by first-order jet interpolation [11, Corollary 1.3]. Before the present work, strong dominability was known to be necessary for the Oka property and to imply its one-dimensional analogue. Alarcón and Forstnerič introduced Oka-1 manifolds and proved, more generally, an Oka-1 principle under dominability away from a closed exceptional set of zero $(2n - 1)$ -dimensional Hausdorff measure [1, Theorem 2.2 and Corollary 2.5]. Thus strong dominability gives approximation and jet interpolation for maps from open Riemann surfaces. Further examples and properties of Oka-1 manifolds were developed by Forstnerič–Lárusson [19].

The construction of new examples has been both a driving force and a source of new ideas in Oka theory: each new class tests which geometric forms of flexibility can be converted into approximation from Stein sources of arbitrary dimension. Two problems have been especially influential. Whether K3 surfaces are Oka was raised by Forstnerič and Lárusson [17, Problem C] and subsequently recorded in Forstnerič’s monograph [15, Problem 7.6.13] and later survey [16, Section 3.3]. Xie and Zhao constructed the first known Oka K3 surfaces, proved that the Oka locus is a dense G_δ subset in every local Kuranishi family, and formulated the conjecture that every complex K3 surface is Oka [34]. The folklore expectation that Fano manifolds are Oka belongs to the broader conjectural picture that smooth projective rationally connected manifolds should be Oka. Alarcón and Forstnerič formulated its Oka-1 analogue [1, Conjecture 9.1].

These questions stimulated several complementary lines of work. Forstnerič and Wold constructed new Oka domains in Euclidean spaces [21]. Xie proved that, for every $n \geq 2$, the complement of a cubic hypersurface $D \subset \mathbb{P}^n$ is Oka unless D_{red} is the union of three distinct hyperplanes containing a common projective linear subspace of codimension two [33, Theorem 1.1]. Du, Guo, and Xie subsequently developed analytic constructions of rational curves on Fano manifolds and Oka-1 approximation for smooth projective morphisms with connected rationally connected fibers [7, 6]. Du, Guo, Wang, and Xie developed periodic-transport constructions for Oka complements [5]; Guo and Xie treated quartic hypersurfaces [24]; and An and Xie treated Fano threefolds [3]. Wang, Xie, and Zhao proved that every K3 surface carrying a genus-one fibration and every generalized Kummer surface is Oka; in particular, every complex Kummer surface is Oka [32].

The present work was directly inspired by OpenAI’s groundbreaking proof that every complex K3 surface is Oka [29]. We use the polynomial source-coordinate and interface-gluing constructions from Sections 3–4 of that paper and extend its Theorem 3.1 from surface targets to targets of arbitrary dimension. The deduction of the local hypothesis from strong dominability and its propagation across proper analytic subsets are established here. Further applications are developed in [4]. In an unpublished companion manuscript, An, Guo, Wang, and Xie use the criterion to prove that every smooth projective rationally connected manifold is Oka [2].

Theorem 1.3. *A complex manifold is Oka if and only if it is strongly dominable.*

Kusakabe [26] proved that a complex manifold covered by Zariski-open Oka subsets is Oka, and asked whether openness in the Euclidean topology would suffice. The following theorem is a direct consequence of Theorem 1.3 and answers this question affirmatively.

Theorem 1.4. *If every point of a complex manifold has an open Oka neighborhood, then the manifold is Oka.*

Theorem 1.4 assumes Oka neighborhoods at every point. If an Oka open subset has analytic complement, the following theorem extends the Oka property across that complement.

Theorem 1.5. *Let X be a complex manifold and $A \subset X$ a proper closed complex analytic subset. If $X \setminus A$ is Oka, then X is Oka.*

In fact, it is enough that X be dominable at every point of $X \setminus A$ by entire maps with values in X ; see Corollary 5.5.

Theorems 1.3 and 1.5 also yield concrete applications. A finite Oka stratification is a finite descending filtration by closed complex subvarieties whose successive differences have Oka connected components [18, Definition 1]. Combining Theorem 1.3 with Forstnerič–Lárusson’s stratification theorem [18, Theorem 2], we obtain that every manifold with a finite Oka stratification is Oka (Corollary 4.5). Xie proved that the space $R_3 = \text{Hol}_3(\mathbb{P}^1, \mathbb{P}^1)$ of cubic rational maps is Oka [33, Theorem 1.3]. More generally, the Euclidean algorithm exhibits a dense Zariski-open Oka cell in the normalized fiber $\mathcal{R}_d = \text{ev}_\infty^{-1}(0)$. Theorem 1.5 makes $\mathcal{R}_d$ Oka, and the evaluation bundle then gives the Oka property of R_d for every $d \geq 1$ (Proposition 6.1).

The proofs of Theorems 1.3 and 1.5 follow a common local-to-global principle. First, a dominating entire map gives the local planar approximation property $\mathcal{L}_n$ of Definition 2.1: a local inverse lifts a map to $\mathbb{C}^n$, where polynomial approximation is available. Second, the source-side constructions of [29], together with gluing and exhaustion, promote this planar property to convex approximation in every source dimension. Third, an expanding-family argument propagates the same local property across a proper analytic subset.

More precisely, the polynomial source coordinates from [29, Section 3] organize transverse parameters so that planar extension may be iterated without losing holomorphic dependence. This gives entire approximation on polydiscs. A real-face induction, using the interface gluing theorem of [29, Section 4], then passes from polydiscs to arbitrary compact convex sets and proves Theorem 1.3.

For Theorem 1.5, a small coordinate line disc centered at a point of the analytic subset may be chosen with boundary in its complement. Planar extension along nearby discs produces expanding holomorphic families in independent tangent directions. An adaptation of the scalar-substitution argument of An–Xie [3, Lemma 4.3] transfers the local property from the boundary to the center and hence across the analytic subset.

This architecture also determines the organization of the paper. Section 2 develops the local approximation and gluing tools. Section 3 globalizes them on polydiscs, and Section 4 completes the passage to convex approximation and proves Theorem 1.3. Section 5 establishes propagation across analytic subsets and proves Theorem 1.5; Section 6 applies the union theorem to spaces of rational functions.

2. LOCAL APPROXIMATION AND GLUING

The local property $\mathcal{L}_n$ connects pointwise domination with approximation of maps. This section establishes that connection and develops the gluing tools needed to extend a planar map while preserving its holomorphic parameters.

For a compact set K , the notation $\text{Op } K$ denotes an unspecified open neighborhood of K , and $\mathcal{O}(\text{Op } K, Y)$ denotes maps to Y holomorphic on such a neighborhood. Let d_Y be a complete distance inducing the topology of the target manifold Y ; its existence follows from the complete Riemannian metric theorem of Nomizu–Ozeki [27]. Completeness will be used in the exhaustion argument of Proposition 3.2.

For $b > 0$, we write

$$\Delta_b = \{\zeta \in \mathbb{C} : |\zeta| < b\}, \quad \Delta = \Delta_1, \quad B_b = \{v \in \mathbb{C}^n : |v| < b\}.$$

Here $|\cdot|$ denotes the Euclidean norm. For a subset E of a complex Euclidean space, $\text{int } E$ denotes its interior in that space; $E \Subset U$ means that $\overline{E}$ is compact and contained in U . The closed polydisc $\overline{\Delta}_b^m$ consists of the points $z \in \mathbb{C}^m$ with $|z_j| \leq b$ for $1 \leq j \leq m$; the balls B_b are reserved for the fiber variables of the target dimension n . In planar approximation, the term *passive parameters* refers to the variables other than the planar variable being extended:

holomorphic dependence on these variables is preserved on a prescribed smaller polydisc. The dimension of each such parameter polydisc is specified at its introduction.

Definition 2.1. A *special planar disc pair* $D \subset D'$ consists of closed topological discs $D, D' \subset \mathbb{C}$ with piecewise smooth boundaries, where D' is obtained from D by attaching another disc along a proper boundary arc. An n -dimensional complex manifold Y has property $\mathcal{L}_n$ at y if there is an open neighborhood $V \ni y$ such that: for every special disc pair $D \subset D'$, integer $q \geq 0$, pair of closed polydiscs

$$P_0 \Subset \text{int } P \subset \mathbb{C}^q$$

and $f \in \mathcal{O}(\text{Op}(D \times P), V)$, the map f can be approximated uniformly on $D \times P_0$ by maps in $\mathcal{O}(\text{Op}(D' \times P_0), Y)$. The set V is a *witness neighborhood* for $\mathcal{L}_n$ at y . A witness neighborhood contained in a target coordinate chart is called a *coordinate witness neighborhood*.

This is property [29, Definition 2.1] in target dimension n , with the inner parameter domain P_0 prescribed before approximation.

Theorem 2.2. *Let Y be an n -dimensional complex manifold. If Y has property $\mathcal{L}_n$ at every point, then Y has CAP in every source dimension.*

The proof extends [29, Theorem 3.1]: polynomial changes of source coordinates make planar extension applicable on a joining region, after which gluing and exhaustion give CAP. Sections 3–4 carry out this passage.

A local inverse lifts the map to $\mathbb{C}^n$. Because the dominating map is entire, polynomial approximation of the lift gives Y -valued approximants without a range restriction.

Lemma 2.3. *If Y is dominable at y , then Y has property $\mathcal{L}_n$ at that point.*

Proof. Choose $H : \mathbb{C}^n \rightarrow Y$ with $H(0) = y$ and dH_0 invertible. The inverse mapping theorem gives neighborhoods $0 \in W \subset \mathbb{C}^n$ and $y \in V \subset Y$ such that $H|_W : W \rightarrow V$ is biholomorphic. For the map f in the definition of the local property, set

$$g = (H|_W)^{-1} \circ f.$$

A closed planar topological disc has connected complement and is therefore polynomially convex; closed polydiscs are polynomially convex, and the product of finitely many polynomially convex compact sets is polynomially convex. The Oka–Weil theorem [9, Theorem 18], applied componentwise, followed by Taylor truncation of the entire approximants, gives polynomial maps

$$g_j : \mathbb{C}^{1+q} \rightarrow \mathbb{C}^n, \quad g_j \rightarrow g \quad \text{uniformly on } D \times P.$$

Choose a compact neighborhood $L \subset \mathbb{C}^n$ of $g(D \times P)$ containing $g_j(D \times P)$ for all large j , and let $\omega_{H,L}$ be a modulus of continuity of $H|_L$. Then

$$F_j = H \circ g_j \in \mathcal{O}(\mathbb{C}^{1+q}, Y), \quad \sup_{D \times P} d_Y(F_j, f) \leq \omega_{H,L}(\|g_j - g\|_{D \times P}) \rightarrow 0.$$

□

Our gluing construction uses holomorphic coordinates for perturbations along the source. When the pulled-back tangent bundle is trivial, a holomorphic frame supplies the fiber variables in the following chart spray. We use the chart-spray and planar-gluing constructions of [29, Lemmas 2.6 and 2.8] in arbitrary target dimension; replacing balls by polydiscs changes only the constants.

Lemma 2.4. *Let $f : S \rightarrow Y$ be holomorphic, where S is Stein, f^*TY is holomorphically trivial on S , and $K \Subset S$. There is a holomorphic map on a product neighborhood of $K \times \overline{\Delta}_\beta^n$,*

$$f^\sharp(z, v), \quad v \in \overline{\Delta}_\beta^n, \quad f^\sharp(z, 0) = f(z),$$

such that each map $v \mapsto f^\sharp(z, v)$ is injective with invertible differential. On smaller compact sets and smaller parameter domains the inverse-chart margins can be chosen uniformly. The triviality hypothesis is satisfied by contractible Stein sources and by products of planar domains with polydiscs.

Proof. The graph $\Gamma_f = \{(z, f(z)) : z \in S\} \subset S \times Y$ is a closed Stein submanifold and hence has a Stein neighborhood by Siu's theorem [31]; this is the case $K = \emptyset$ of [11, Theorem 2.1]. Let $\mathcal{N}$ be this Stein neighborhood, $\mathcal{I}_{\Gamma_f}$ its graph ideal, and $\mathcal{E}$ the sheaf of holomorphic sections of the vertical bundle $\pi_Y^* TY$. Cartan's theorem B [30, Theorem B], applied to the coherent sheaf $\mathcal{I}_{\Gamma_f} \mathcal{E}$ on $\mathcal{N}$, gives

$$(2.1) \quad 0 \longrightarrow \mathcal{I}_{\Gamma_f} \mathcal{E} \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}|_{\Gamma_f} \longrightarrow 0, \quad H^1(\mathcal{N}, \mathcal{I}_{\Gamma_f} \mathcal{E}) = 0.$$

The vanishing in (2.1) makes the restriction map on global sections surjective. A frame $e_1, \dots, e_n$ of $f^* TY$ has vertical holomorphic extensions V_j with

$$V_j(z, f(z)) = e_j(z), \quad 1 \leq j \leq n.$$

The fields $V_1, \dots, V_n$ are vertical and extend the frame. Writing ϕ_j^t for the local flow of V_j and $\varepsilon_1, \dots, \varepsilon_n$ for the standard basis of $\mathbb{C}^n$, we obtain

$$f^\sharp(z, v) = \pi_Y(\phi_n^{v_n} \circ \dots \circ \phi_1^{v_1}(z, f(z))), \quad D_v f^\sharp(z, 0) \varepsilon_j = V_j(z, f(z)).$$

For the fiber-preserving map $\Psi(z, v) = (z, f^\sharp(z, v))$,

$$(2.2) \quad D\Psi_{(z,0)} = \begin{pmatrix} I & 0 \\ D_z f & D_v f^\sharp(z, 0) \end{pmatrix}, \quad \det D_v f^\sharp(z, 0) \neq 0.$$

By (2.2), the inverse function theorem gives product neighborhoods $U_i \times \Delta_{\beta_i}^n$ on which Ψ is injective and has a holomorphic inverse. A finite subcover of K and

$$(2.3) \quad 0 < \beta < \min_i \beta_i, \quad K \times \overline{\Delta}_\beta^n \Subset \bigcup_i U_i \times \Delta_{\beta_i}^n$$

give a common fiber radius by (2.3). The inverse neighborhoods of the compact images of smaller products give the asserted uniform margins.

Complex vector bundles over a contractible source are topologically trivial. A planar domain has the homotopy type of a CW complex of dimension at most one [16, Section 1], and $\mathrm{GL}_n(\mathbb{C})$ is connected, so a frame over the vertices extends over every edge. Thus complex vector bundles over the planar domain, and over its product with a polydisc, are topologically trivial. The Oka–Grauert classification of bundles [9, Section 7.1] then gives holomorphic triviality. $\square$

In these fiber coordinates, the linear step in gluing is an additive Cousin problem on the overlap. The following planar splitting, from [29, Lemma 2.4], solves it with bounds independent of the number of holomorphic parameters.

For an open set $U \subset \mathbb{C}^m$, let

$$H^\infty(U, \mathbb{C}^N) = \{f \in \mathcal{O}(U, \mathbb{C}^N) : \sup_{z \in U} |f(z)| < \infty\}$$

be the Banach space of bounded holomorphic maps, equipped with the supremum norm $\|f\|_{\infty, U} = \sup_{z \in U} |f(z)|$.

Lemma 2.5. *Suppose the bounded planar open sets U_A, U_B have separated closed fringes,*

$$\mathrm{dist}(\overline{U_A} \setminus U_B, \overline{U_B} \setminus U_A) > 0.$$

For their overlap $C = U_A \cap U_B$ and every passive parameter polydisc P , there are bounded linear operators R_A, R_B such that

$$(2.4) \quad R_A u - R_B u = u, \quad \|R_A u\|_{U_A \times P} + \|R_B u\|_{U_B \times P} \leq C_0 \|u\|_{C \times P}.$$

They act on bounded holomorphic functions with values in any finite-dimensional vector space and preserve holomorphic dependence on all passive parameters. The constant C_0 is independent of the radii and of the number of passive parameters.

For a smooth planar cutoff χ , equal to zero near $\overline{U_A \setminus U_B}$ and to one near $\overline{U_B \setminus U_A}$, the operators of [29, proof of Lemma 2.4] satisfy the quantitative bound

$$(2.5) \quad C_0 = 2 + 4d \|\partial_{\bar{s}} \chi\|_{\infty}, \quad d \geq \text{diam}(U_A \cup U_B).$$

The operators in (2.4) give the planar spray gluing of [29, Lemma 2.8]: a contraction converts additive splitting into exact agreement of target-valued maps. Suppose two sprays F_A, F_B are sufficiently close on the overlap, the first has uniform inverse-chart margins, and the auxiliary parameter polydisc has a positive reserved margin. Write

$$(2.6) \quad \gamma(s, t, v) = F_A(s, t, \cdot)^{-1} \circ F_B(s, t, v) = v + E(s, t, v).$$

Reserve a fiber polydisc Δ_{ρ}^n and a smaller passive parameter domain. In the maximum norm, let

$$(2.7) \quad e = \|E\|_{\infty}, \quad \kappa = \|D_v E\|_{\infty}, \quad 2C_0 e < \rho, \quad C_0 \kappa < \frac{1}{2}.$$

The smallness conditions in (2.7) are met once F_B approximates F_A closely on a larger fiber domain, by Cauchy estimates. On the ball $\|u\| \leq 2e$ in $H^{\infty}((U_A \cap U_B) \times P, \mathbb{C}^n)$, the operator

$$(2.8) \quad \mathcal{T}(u)(s, t) = E(s, t, R_B u(s, t))$$

satisfies

$$(2.9) \quad \|\mathcal{T}(u)\| \leq e, \quad \|\mathcal{T}(u) - \mathcal{T}(\tilde{u})\| \leq C_0 \kappa \|u - \tilde{u}\|.$$

By (2.9), the fixed point of (2.8) satisfies $\|u\| \leq e$. With $a = R_A u$ and $b = R_B u$,

$$(2.10) \quad a - b = u = E(s, t, b), \quad a = \gamma(s, t, b), \quad \|a\| + \|b\| \leq C_0 e.$$

The corrections in (2.10) give the holomorphic map

$$F(s, t) = \begin{cases} F_A(s, t, a(s, t)), & s \in U_A, \\ F_B(s, t, b(s, t)), & s \in U_B, \end{cases}$$

By (2.6) and (2.10), $F_A(s, t, a) = F_B(s, t, b)$ on $U_A \cap U_B$. For a fixed compact $K_A \subset U_A \times P$ and a continuity modulus ω_{A, K_A} of F_A in its fiber variable,

$$\sup_{K_A} d_Y(F, F_A(\cdot, 0)) \leq \omega_{A, K_A}(C_0 e).$$

The bound (2.5) depends only on d and $\|\partial_{\bar{s}} \chi\|_{\infty}$; these quantities are specified in the planar applications below.

We now enlarge the planar domain while retaining any prescribed smaller parameter polydisc. The disc case will be used to construct expanding families in Section 5; the annular case will extend the maps expressed in the source coordinates of Section 3.

Proposition 2.6. *Let $r > 0$,*

$$K = \overline{\Delta}_r \quad \text{or} \quad K = \{s \in \mathbb{C} : r_- \leq |s| \leq r\}, \quad 0 < r_- < r,$$

and suppose that f is holomorphic near $K \times P$ for a closed polydisc $P \subset \mathbb{C}^q$. Suppose in addition that for every $|\zeta| = r$ the set $f(\{\zeta\} \times P)$ lies in a coordinate witness neighborhood. For every $R > r$, prescribed $P_0 \Subset \text{int } P$, and $\epsilon > 0$, there is

$$F \in \mathcal{O}(\text{Op}(K_R \times P_0), Y), \quad \sup_{K \times P_0} d_Y(F, f) < \epsilon,$$

where $K_R = \overline{\Delta}_R$ in the disc case and $K_R = \{r_- \leq |s| \leq R\}$ in the annular case.

Proof. We follow the proof of [29, Proposition 2.9]. The image of the outer circle has a finite cover by the witness neighborhoods: choose $P_0 \Subset \text{int } \hat{P} \Subset \text{int } P$ and a finite cyclic cover of the outer circle by slightly enlarged arcs $I_1, \dots, I_\ell$ such that there are coordinate witness neighborhoods $V_1, \dots, V_\ell$ with

$$f(I_j \times \hat{P}) \Subset V_j.$$

Radial extension. The proof glues in $N = \ell + 1$ operations: one radial gluing followed by one operation for each sector. Choose a chain of parameter domains and positive errors with

$$(2.11) \quad P_0 \Subset \text{int } P^{(N)} \Subset \dots \Subset \text{int } P^{(0)} \Subset \text{int } \hat{P}, \quad \epsilon_k > 0, \quad \sum_{k=1}^N \epsilon_k < \epsilon.$$

The intermediate polydiscs $\hat{P}$ and $P^{(k)}$ are chosen concentric with P ; we translate the parameter coordinates so that their common center is 0. Choose division points $\zeta_j \in I_j \cap I_{j+1}$, with cyclic indices, so that the arc from ζ_{j-1} to ζ_j lies in I_j and has angular width less than π . Then

$$f(\{\zeta_j\} \times \hat{P}) \Subset V_j \cap V_{j+1}.$$

By Lemma 2.4, choose a chart spray $f^\sharp$ near $K \times \hat{P}$; the planar source times a polydisc satisfies its triviality hypothesis. In applying $\mathcal{L}_n$ to this spray, the original parameters $t \in \mathbb{C}^q$ and the fiber variables $v \in \mathbb{C}^n$ together form the parameter tuple $(t, v) \in \mathbb{C}^{q+n}$. Only the planar domain is enlarged. After reducing its fiber radius, choose mutually disjoint thin filled polar rectangles T_j crossing the outer circle at ζ_j so that

$$f^\sharp(T_j \times \hat{P} \times \overline{\Delta}_\beta^n) \Subset V_j \cap V_{j+1}.$$

Attach to the outer edge of T_j a radial segment L_j ending beyond radius R , and denote its attachment point by a_j . Set

$$E_j = T_j \cup L_j, \quad w = (t, v).$$

In a target coordinate chart ψ_j on V_j , extend the spray coordinates by

$$u_j(s, w) = \begin{cases} \psi_j(f^\sharp(s, w)), & s \in T_j, \\ \psi_j(f^\sharp(a_j, w)), & s \in L_j. \end{cases}$$

The set E_j has connected complement. For a coordinate tolerance $\delta > 0$, take a Taylor truncation on the smaller parameter polydisc $W = P^{(0)} \times \overline{\Delta}_{\beta/2}^n$ such that

$$(2.12) \quad T_{j,M}(s, w) = \sum_{|\nu| \leq M} a_{j,\nu}(s) w^\nu, \quad \|u_j - T_{j,M}\|_{E_j \times W} < \delta/2.$$

Each coefficient is continuous on E_j and holomorphic on its interior. Mergelyan's theorem [9, Theorem 3], applied componentwise, supplies polynomials $p_{j,\nu}$ such that

$$(2.13) \quad \sum_{|\nu| \leq M} \|a_{j,\nu} - p_{j,\nu}\|_{E_j} \sup_{w \in W} |w^\nu| < \delta/2.$$

Combining (2.12) and (2.13) gives

$$(2.14) \quad p_j(s, w) = \sum_{|\nu| \leq M} p_{j,\nu}(s) w^\nu, \quad \|u_j - p_j\|_{E_j \times W} < \delta.$$

The compact coordinate image has a positive margin inside $\psi_j(V_j \cap V_{j+1})$. Thus, by (2.14), if $\|u_j - p_j\|_{E_j \times W}$ is less than half that margin, then p_j maps $E_j \times W$ into $\psi_j(V_j \cap V_{j+1})$, and the same holds on $U_j \times W$ for $U_j \supset E_j$ thin enough that

$$(2.15) \quad G_j = \psi_j^{-1} \circ p_j \in \mathcal{O}(U_j \times W, V_j \cap V_{j+1}), \quad \sup_{T_j \times W} d_Y(G_j, f^\sharp) < \eta_j.$$

Let Γ_j be the radial segment from ζ_j to the outer endpoint of L_j . Its initial part lies in T_j . Choose fixed radii $r < a < b$ inside the initial parts of the radial segments, with

$$\Gamma_j \cap \{a \leq |s| \leq b\} \subseteq \text{int } T_j.$$

Take a bounded open neighborhood O of $K \cup \bigcup_j \Gamma_j$ such that

$$(2.16) \quad \begin{aligned} \overline{O \cap \{a \leq |s| \leq b\}} &\subset \bigcup_j \text{int } T_j, \\ O \cap \{|s| < b\} &\subset \text{dom}_s f^\sharp, \\ O \cap \{|s| > a\} &\subset \bigcup_j U_j. \end{aligned}$$

The source domain of $f^\sharp$ is taken with the fixed parameter product W , and the neighborhoods U_j are pairwise disjoint. By (2.16), the two sprays and the planar domains on which they are defined are

$$A = O \cap \{|s| < b\}, \quad B = O \cap \{|s| > a\}, \quad F_A = f^\sharp, \quad F_B|_{B \cap U_j} = G_j.$$

Consequently

$$\begin{aligned} A \cup B &= O, \quad A \cap B = O \cap \{a < |s| < b\}, \quad K \subset A, \\ \text{dist}(\overline{A \setminus B}, \overline{B \setminus A}) &\geq b - a. \end{aligned}$$

Choose a fixed $\vartheta \in C^\infty(\mathbb{R}, [0, 1])$ with $\vartheta = 0$ on $(-\infty, 1/3]$ and $\vartheta = 1$ on $[2/3, \infty)$, and set

$$(2.17) \quad \chi(s) = \vartheta\left(\frac{|s| - a}{b - a}\right), \quad \|\partial_{\bar{s}} \chi\|_\infty \leq \frac{\|\vartheta'\|_\infty}{2(b - a)}.$$

The value of χ is constant near $s = 0$, so χ is smooth on $\mathbb{C}$. All domains lie in a fixed disc of diameter at most D_* . Equations (2.5) and (2.17) give operators with

$$(2.18) \quad \|R_A\| + \|R_B\| \leq 2 + \frac{2D_* \|\vartheta'\|_\infty}{b - a} =: C_*.$$

The radii a, b , the cutoff, and the disc of diameter D_* are fixed before the neighborhoods U_j are narrowed outside the rectangles.

On the overlap and a smaller fiber polydisc, write

$$(2.19) \quad \gamma_0(s, t, v) = f^\sharp(s, t, \cdot)^{-1}(F_B(s, t, v)) = v + c_0(s, t, v).$$

Equation (2.15) and Cauchy estimates make $c_0, D_v c_0$ in (2.19) satisfy (2.7), with $C_0 = C_*$ from (2.18), once the approximation errors η_j are small. Solve

$$(2.20) \quad u_0 = c_0(s, t, R_B u_0), \quad a_0 = R_A u_0, \quad b_0 = R_B u_0.$$

For t near $P^{(1)}$, the resulting map and its distance to f satisfy

$$(2.21) \quad \begin{aligned} F_0(s, t) &= \begin{cases} f^\sharp(s, t, a_0(s, t)), & s \in A, \\ G_j(s, t, b_0(s, t)), & s \in B \cap U_j, \end{cases} \\ \sup_{K \times P^{(1)}} d_Y(F_0, f) &\leq \omega_{f^\sharp, K \times P^{(1)}}(C_* \|c_0\|_\infty) < \epsilon_1. \end{aligned}$$

Equation (2.20) gives $a_0 = \gamma_0(s, t, b_0)$, so the two expressions in (2.21) agree on $A \cap B$. Choose an outer radius $R_* > R$ below the radius of the outer endpoint of every spoke Γ_j ; each spoke therefore meets the circle $\{|s| = R_*\}$. Choose thin polar rectangular neighborhoods U_j^0 of the spokes with

$$U_j^0 \subseteq O \cap U_j, \quad \Gamma_j \cap \{|s| \leq R_*\} \subset U_j^0.$$

They are pairwise disjoint and thin enough that each lies in its own angular wedge. After a cyclic relabeling of the spokes we may also assume that U_{j-1}^0 and U_j^0 bound the sector S_j

introduced below. For small corrections their images remain in $V_j \cap V_{j+1}$, and F_0 is holomorphic near the compact set

$$K_0 = K \cup \bigcup_{j=1}^{\ell} \overline{U_j^0}.$$

Filling the sectors. Let S_j , $1 \leq j \leq \ell$, be the closed polar sector between Γ_{j-1} and Γ_j , of radial range $r \leq |s| \leq R_*$, with indices read cyclically. Write

$$K_j = K_0 \cup S_1 \cup \cdots \cup S_j;$$

then $S_i \cap S_j \subset K_0$ for $i \neq j$, $S_i \cap K_0 = J_i$, and $K_R \subset K_\ell$. Distinct sectors therefore meet only along radial edges contained in K_0 , and the unfilled sectors have the fixed joining sets

$$J_j = K_0 \cap S_j = K_{j-1} \cap S_j.$$

The base arc of J_j lies in I_j and its radial portions lie in $U_{j-1}^0 \cup U_j^0$. Taking the initial corrections sufficiently small gives $F_0(J_j \times P^{(1)}) \Subset V_j$ for every j , and the error bounds (2.29) preserve these chart containments. Choose a thin U-shaped closed band D_j around J_j and a larger disc D'_j such that

$$\begin{aligned} S_j \cap K_{j-1} &\subset \text{int } D_j, & D_j &\subset \text{dom}_s F_{j-1}, \\ S_j &\subset \text{int } D'_j, & D_j &\subset D'_j \text{ is a special disc pair.} \end{aligned}$$

Compactness allows the band to be chosen with its entire current parameter image in V_j . A chart spray for F_{j-1} near K_{j-1} then satisfies, after reducing its fiber radius,

$$F_{j-1}^\#(D_j \times P^{(k-1)} \times \overline{\Delta}_{\beta_j}^n) \Subset V_j.$$

Here $k = j + 1$ counts the initial radial gluing and the subsequent sector gluings. Prescribe the retained original and spray parameter domains

$$W_j = P^{(k-1)} \times \overline{\Delta}_{\beta_j}^n, \quad W'_j = P^{(k)} \times \overline{\Delta}_{\beta'_j}^n, \quad W'_j \Subset \text{int } W_j, \quad 0 < \beta'_j < \beta_j,$$

before applying $\mathcal{L}_n$. Property $\mathcal{L}_n$ then gives a holomorphic map H_j near $D'_j \times W'_j$ with

$$(2.22) \quad \sup_{D_j \times W'_j} d_Y(H_j, F_{j-1}^\#) < \eta_j.$$

Choose planar open bands $\mathcal{W}_j^0 \Subset \mathcal{W}_j^1 \Subset \text{int } D_j$ containing $S_j \cap K_{j-1}$. The compact remainders

$$E_j^- = K_{j-1} \setminus \mathcal{W}_j^0, \quad E_j^+ = S_j \setminus \mathcal{W}_j^0$$

are disjoint. Choose $\lambda_j \in C_c^\infty(\mathbb{C}, [0, 1])$ equal to zero near E_j^- and to one near E_j^+ . Shrink a bounded neighborhood O_j of K_j so that

$$O_j \cap \{1/3 \leq \lambda_j \leq 2/3\} \Subset \mathcal{W}_j^1.$$

The two domains and their overlap are

$$(2.23) \quad \begin{aligned} A_j &= O_j \cap \{\lambda_j < 2/3\} \subset \text{dom}_s F_{j-1}^\#, \\ B_j &= O_j \cap \{\lambda_j > 1/3\} \subset \text{dom}_s H_j, \\ A_j \cup B_j &= O_j, \quad A_j \cap B_j \Subset \text{int } D_j, \quad B_j \cap K_{j-1} \subset \mathcal{W}_j^0. \end{aligned}$$

The inclusions in (2.23) hold because the corresponding compact portions of K_j lie in the old domain or in $\text{int } D'_j$, respectively. Choose $\theta \in C^\infty(\mathbb{R}, [0, 1])$ equal to zero on $(-\infty, 2/5]$ and to one on $[3/5, \infty)$. For $\chi_j = \theta \circ \lambda_j$,

$$(2.24) \quad \begin{aligned} \text{dist}(\overline{A_j \setminus B_j}, \overline{B_j \setminus A_j}) &\geq \frac{1}{3 \|\nabla \lambda_j\|_\infty}, \\ \|\nabla \chi_j\|_\infty &\leq \|\theta'\|_\infty \|\nabla \lambda_j\|_\infty. \end{aligned}$$

By (2.5) and (2.24), Lemma 2.5 supplies bounded operators R_{A_j}, R_{B_j} with a fixed bound C_j .

On the overlap, the transition and its corrections satisfy

$$\begin{aligned}\gamma_j &= (F_{j-1}^\sharp)^{-1} \circ H_j = \text{id} + c_j, \\ u_j &= c_j(s, t, R_{B_j} u_j), \quad a_j = R_{A_j} u_j, \quad b_j = R_{B_j} u_j,\end{aligned}$$

where inverses and compositions act in the fiber variable. For small η_j ,

$$(2.25) \quad a_j = \gamma_j(s, t, b_j), \quad \|a_j\| + \|b_j\| \leq C_j \|c_j\|_\infty.$$

The corrections (2.25) give the map near $K_j \times P^{(k)}$

$$(2.26) \quad F_j(s, t) = \begin{cases} F_{j-1}^\sharp(s, t, a_j(s, t)), & s \in A_j, \\ H_j(s, t, b_j(s, t)), & s \in B_j. \end{cases}$$

Equation (2.25) makes the expressions in (2.26) agree on $A_j \cap B_j$. On $B_j \cap K_{j-1}$, (2.23) makes (2.22) applicable. The continuity modulus ω_j of the fixed old spray $F_{j-1}^\sharp$ therefore gives

$$(2.27) \quad \sup_{K_{j-1} \times P^{(k)}} d_Y(F_j, F_{j-1}) \leq \eta_j + \omega_j(C_j \|c_j\|_\infty) < \epsilon_k.$$

Chart preservation and the final error. At each operation the n spray variables join the original q passive parameters. During a sector operation $k \geq 2$, the sectors that are still unfilled have the fixed compact joining sets J_j , $j \geq k-1$, with current chart margins

$$(2.28) \quad \mu_k = \min_{j \geq k-1} \text{dist}(F_{\text{old}}(J_j \times P^{(k-1)}), Y \setminus V_j) > 0.$$

Once the overlap, the cutoff, and the spray radius are fixed, η_j can be chosen so that

$$(2.29) \quad \sup_{K_{\text{old}} \times P^{(k)}} d_Y(F_{\text{new}}, F_{\text{old}}) < \min(\epsilon_k, \mu_k/2).$$

Equations (2.28) and (2.29) give $F_{\text{new}}(J_j \times P^{(k)}) \subset V_j$. Since $K_{\text{new}} \cap S_j = J_j$ for every remaining sector, the chart domains required by the next operation are still available. The two sources of error are the initial gluing error in (2.21) and the sector gluing errors in (2.27), and by (2.11) their sum is bounded by $\sum_{k=1}^N \epsilon_k < \epsilon$. The final map F is holomorphic near $K_R \times P_0$, and (2.11), (2.21), and (2.27) yield

$$\sup_{K \times P_0} d_Y(F, f) \leq \sum_{k=1}^N \epsilon_k < \epsilon.$$

□

3. POLYNOMIAL SOURCE COORDINATES AND POLYDISC APPROXIMATION

To apply planar extension in several variables, the image of the transverse parameter polydisc over each planar point in the joining region must lie in one witness neighborhood. The polynomial source coordinates of [29, Lemma 3.3] achieve this by making the transverse fibers small in the original source coordinates. We retain their formulas to track the estimates needed for gluing.

Lemma 3.1. *Fix $m \geq 2$ and*

$$1 < r_m < \cdots < r_2 < r_1, \quad 0 < a < 1, \quad r_j < ar_{j-1} \quad (2 \leq j \leq m).$$

Set

$$P_j(w) = (w/r_j)^d, \quad H_d(x) = (ax_m, P_2(x_2) - ax_1, \dots, P_m(x_m) - ax_{m-1}).$$

There exist $C \geq 1$, $q \in (0, 1)$, and an integer $d_0 \geq 2$, depending only on the fixed data, such that for every pair of integers $d \geq d_0$ and $J \geq 1$, the set $E_{d,J} = H_d^{-J}(\overline{\Delta}_1^m)$ satisfies

$$(3.1) \quad \max_{j \geq 2} |P_j(x_j)| \leq C \max(1, |x_1|).$$

If also $|x_1| > r_1$, then all x_j are nonzero and

$$(3.2) \quad \left| \frac{H_d(x)_j}{ax_{j-1}} \right| \leq q \quad (2 \leq j \leq m).$$

By (3.2), the numbers

$$\xi_j = \log(P_j(x_j)/(ax_{j-1})) \quad (2 \leq j \leq m)$$

are defined using the same logarithm branch centered at 1, and satisfy $|\xi_j| \leq B = -\log(1-q)$. With

$$D = d^{m-1}, \quad \rho_1 = 1, \quad \rho_j = r_j(a\rho_{j-1})^{1/d},$$

there is a single-valued function s , holomorphic on an open neighborhood of $E_{d,J} \cap \{|x_1| > r_1\}$, such that

$$(3.3) \quad x_j = \rho_j s^{d^{m-j}} \exp\left(\sum_{k=2}^j \frac{\xi_k}{d^{j-k+1}}\right), \quad x_1 = s^D.$$

The polynomial inverse of H_d is obtained, for $y = H_d(x)$, by the recursion

$$x_m = y_1/a, \quad x_{j-1} = (P_j(x_j) - y_j)/a \quad (j = m, m-1, \dots, 2).$$

The single-valued coordinate in (3.3) is

$$s = \frac{x_m}{\rho_m} \exp\left(-\sum_{k=2}^m \frac{\xi_k}{d^{m-k+1}}\right).$$

The next estimates quantify the variation in the transverse parameters on the joining annulus. Write $\Phi_d(s, \xi)$ for the right side of (3.3), now allowing ξ to vary independently, and fix

$$r_1 < c_0 < c_1 < c_2 < c_3 < r_1 + \ell$$

and $\Xi_i = \{\xi : |\xi_j| \leq B + i + 1\}$, $i = 1, 2$. On the annulus $c_0 \leq |s|^D \leq c_3$ and for $\xi \in \Xi_2$ there are fixed constants C_1, C_2 such that

$$(3.4) \quad |\Phi_{d,j}(s, \xi)| \leq r_j e^{C_1/d} \quad (j \geq 2),$$

$$(3.5) \quad \sup_{\xi, \tilde{\xi} \in \Xi_2} \left\| \Phi_d(s, \xi) - \Phi_d(s, \tilde{\xi}) \right\|_{\infty} \leq C_2/d.$$

Equations (3.4) and (3.5) are the estimates in [29, equation (3.8)], with the transverse polydisc fixed at Ξ_2 .

The change of source coordinates is also close to the identity near the original polydisc. With

$$L(x) = (ax_m, -ax_1, \dots, -ax_{m-1}),$$

first fix J with $a^J R_+ < 1$ for a prescribed $R_+ > 1$, and set

$$(3.6) \quad h_d = H_d^{-J} L^J.$$

On a fixed neighborhood of the unit polydisc, $h_d \rightarrow \text{id}$. More precisely, choose $1 < b_0 < b_1 < r_m$ and $\theta = b_1/r_m < 1$. The estimate proving [29, equation (3.7)] gives, for large d ,

$$(3.7) \quad \|h_d - \text{id}\|_{\Delta_{b_0}^m} \leq \theta^d \sum_{i=1}^J a^{-i} \rightarrow 0.$$

The constant J is fixed before d tends to infinity.

Combining these coordinates with planar extension gives the polydisc approximation needed to start the convex-set induction in Section 4.

Proposition 3.2. *If Y satisfies $\mathcal{L}_n$ at every point, then every holomorphic map into Y defined near a closed polydisc can be approximated uniformly on that polydisc by maps entire in all source variables.*

Proof. We first extend to any prescribed finite radius, then use summable approximation errors to obtain an entire limit. Let f be holomorphic near $\overline{\Delta}_1^m$, and fix $R > 1$ and $\epsilon > 0$. For $m = 1$, apply Proposition 2.6 without passive parameters.

For $m \geq 2$, choose $b > 1$ such that f is holomorphic near $\overline{\Delta}_b^m$. Choose data of Lemma 3.1 and $\ell > 0$ with $r_1 + 3\ell < b$. Fix $r_1 + 2\ell < b' < b$, and construct a chart spray $f^\sharp(x, v)$ on a neighborhood of $\overline{\Delta}_{b'}^m \times \overline{\Delta}_\beta^n$. Choose $R_+ > R$, and first fix J in (3.6).

A finite witness cover of the compact set $f(\overline{\Delta}_{b'}^m)$ gives $\delta_0 > 0$ such that, for every $x \in \overline{\Delta}_{b'}^m$, some witness neighborhood contains the target ball $B_Y(f(x), 3\delta_0)$. Reduce the spray radius so that

$$\sup_{x \in \overline{\Delta}_{b'}^m, v \in \overline{\Delta}_\beta^n} d_Y(f^\sharp(x, v), f(x)) < \delta_0.$$

By (3.4), the points $\Phi_d(s, \xi)$ lie in $\overline{\Delta}_{b'}^m$ for large d ; by (3.5) and a continuity modulus ω_f ,

$$(3.8) \quad \begin{aligned} & \sup_{\xi \in \Xi_2, v \in \overline{\Delta}_\beta^n} d_Y(f^\sharp(\Phi_d(s, \xi), v), f(\Phi_d(s, 0))) \\ & \leq \delta_0 + \omega_f(C_2/d) < 2\delta_0, \quad |s|^D = c_3. \end{aligned}$$

Equation (3.8) places the whole parameter image over each such s in one witness neighborhood. Using (3.7), also require

$$(3.9) \quad \sup_{z \in \overline{\Delta}_1^m} d_Y(f(h_d(z)), f(z)) < \epsilon/2, \quad \sup_{z \in \overline{\Delta}_1^m} |h_d(z)_1| < c_1, \quad r_j(Cc_2)^{1/d} < b' \quad (j \geq 2).$$

Now fix $d, h = h_d$, the source geometry, the parameter radii, and all cutoff constants.

Apply Proposition 2.6 to the map

$$(s, \xi, v) \longmapsto f^\sharp(\Phi_d(s, \xi), v),$$

which is holomorphic near the annulus $\{c_0 \leq |s|^D \leq c_3\} \times \Xi_2 \times \overline{\Delta}_\beta^n$ for large d by (3.4); treat both ξ and v as passive parameters. Prescribe the inner parameter domain $\Xi_1 \times \overline{\Delta}_{\beta/2}^n$ and prescribe $\eta > 0$. This gives a spray $G(s, \xi, v)$ with

$$(3.10) \quad \sup_{\{c_0 \leq |s|^D \leq c_3\} \times \Xi_1 \times \overline{\Delta}_{\beta/2}^n} d_Y(G, f^\sharp(\Phi_d(\cdot, \cdot), \cdot)) < \eta,$$

defined out to any further prescribed finite outer radius. Choose that radius larger than both $c_3^{1/D}$ and

$$(3.11) \quad \sup\{|s(h(z))| : z \in \overline{\Delta}_{R_+}^m, |h(z)_1| \geq c_1\}.$$

If the set is empty, the second radius requirement is vacuous. Otherwise the supremum in (3.11) is finite by compactness and the single-valued coordinate s in (3.3). The inner radius can be kept below $c_1^{1/D}$.

Since $a^J R_+ < 1$, (3.6) gives $h(\overline{\Delta}_{R_+}^m) \subset E_{d,J}$. Choose a fixed convex neighborhood Ω of $\overline{\Delta}_R^m$ contained in $\overline{\Delta}_{R_+}^m$, and set

$$A = \{z \in \Omega : |h(z)_1| < c_2\}, \quad B' = \{z \in \Omega : |h(z)_1| > c_1\}.$$

Write $x = h(z)$. On the old side, (3.1) gives

$$z \in A \implies |x_1| < c_2 < b', \quad |x_j| \leq r_j(Cc_2)^{1/d} < b' \quad (j \geq 2).$$

On the new side, the end coordinates satisfy

$$(3.12) \quad z \in B' \implies |s(x)|^D = |x_1| > c_1, \quad |\xi_j(x)| \leq B < B + 2.$$

The radius condition (3.11) and the bounds (3.12) place $(s(x), \xi(x))$ inside the domain of G , so the two sprays

$$(3.13) \quad F_A(z, v) = f^\sharp(h(z), v), \quad F_{B'}(z, v) = G(s(h(z)), \xi(h(z)), v)$$

are defined on A and B' , respectively. On $A \cap B'$, one has $c_1 < |s(h(z))|^D < c_2$ and

$$(3.14) \quad \Phi_d(s(h(z)), \xi(h(z))) = h(z),$$

so (3.14) and (3.10) make the two sprays (3.13) close on the fixed overlap and retained fiber domain. On a smaller fixed fiber domain, the inverse charts of F_A give

$$\gamma(z, v) = F_A(z, \cdot)^{-1} \circ F_{B'}(z, v) = v + c(z, v),$$

and $\|c\|_\infty \rightarrow 0$ as the planar approximation error in (3.10) tends to zero.

We now glue the two sprays by Forstnerič's compositional splitting theorem for a separated pair of compact sets whose union has a basis of Stein neighborhoods.

Take a compact convex set K and numbers d_1, d_2 with

$$(3.15) \quad \begin{aligned} \overline{\Delta}_R^m &\Subset \text{int } K \Subset \Omega, & c_1 < d_1 < d_2 < c_2, \\ K_A &= K \cap \{|h_1| \leq d_2\}, & K_B &= K \cap \{|h_1| \geq d_1\}. \end{aligned}$$

Then $K_A \cup K_B = K$ and

$$(3.16) \quad \overline{K_A \setminus K_B} \cap \overline{K_B \setminus K_A} = \emptyset, \quad K_A \cap K_B \Subset A \cap B'.$$

Choose $0 < \rho < \rho_+$ inside the fixed fiber domain of γ . Cauchy estimates give

$$(3.17) \quad \sup_{(K_A \cap K_B) \times \overline{B}_\rho} \|D_v c\| \leq C(\rho_+ - \rho)^{-1} \|c\|_\infty \rightarrow 0.$$

Since the left side of (3.17) tends to 0, it is less than $1/2$ on a neighborhood of $K_A \cap K_B$ for accurate planar approximation. Hence

$$(3.18) \quad \Gamma(z, v) = (z, \gamma(z, v)), \quad |\gamma(z, v) - \gamma(z, w)| \geq \frac{1}{2}|v - w|$$

on a fixed smaller fiber ball. By (3.18), after reducing ρ if necessary, Γ is biholomorphic onto its image near $(K_A \cap K_B) \times \overline{B}_\rho$.

By (3.16), the compact sets $K_A \times \overline{B}_\rho$ and $K_B \times \overline{B}_\rho$ have separated closed fringes. Their union $K \times \overline{B}_\rho$ is convex and therefore has a basis of Stein neighborhoods. Hence the ordered pair $(K_B \times \overline{B}_\rho, K_A \times \overline{B}_\rho)$ satisfies the hypotheses of Forstnerič's compositional splitting theorem [10, Theorem 4.1]. Applying its foliation-preserving form to Γ , with the foliation $\{z = \text{constant}\}$, yields neighborhoods $U_A \supset K_A$, $U_B \supset K_B$ and, on a smaller fixed fiber ball,

$$(3.19) \quad \gamma(z, \alpha_B(z, v)) = \alpha_A(z, v), \quad \|\alpha_A - \text{id}\|_\infty + \|\alpha_B - \text{id}\|_\infty < \eta.$$

The neighborhoods can be restricted to $U_A \Subset A$, $U_B \Subset B'$, and for small η the corrections take values in the original fiber domains. Thus

$$\tilde{f}(z) = \begin{cases} f^\sharp(h(z), \alpha_A(z, 0)), & z \in U_A, \\ G(s(h(z)), \xi(h(z)), \alpha_B(z, 0)), & z \in U_B \end{cases}$$

is holomorphic near $\overline{\Delta}_R^m$, since equality on $U_A \cap U_B$ follows from (3.19). By (3.9) and (3.15), $\overline{\Delta}_1^m \subset K_A$. A continuity modulus of the fixed spray F_A gives

$$\sup_{\overline{\Delta}_1^m} d_Y(\tilde{f}, f) \leq \omega_{A, \overline{\Delta}_1^m}(\eta) + \sup_{\overline{\Delta}_1^m} d_Y(f \circ h, f) < \epsilon$$

once η is chosen with $\omega_{A, \overline{\Delta}_1^m}(\eta) < \epsilon/2$ and the second term is below $\epsilon/2$ by (3.9).

Starting with the original map $f_0 = f$, repeat finite extension over radii $1 = R_0 < R_1 < \dots \rightarrow \infty$, with maps f_j holomorphic near $\overline{\Delta}_{R_j}^m$ and

$$(3.20) \quad \sup_{\overline{\Delta}_{R_{j-1}}^m} d_Y(f_j, f_{j-1}) < \epsilon 2^{-j-1} \quad (j \geq 1).$$

Summing (3.20) for $k > j \geq l$ gives

$$(3.21) \quad \sup_{\overline{\Delta}_{R_l}^m} d_Y(f_k, f_j) < \sum_{\nu=j+1}^k \epsilon 2^{-\nu-1} < \epsilon 2^{-j-1}.$$

Equation (3.21) and completeness of d_Y give a continuous limit F , uniformly on every compact set, with

$$\sup_{\overline{\Delta}_1^m} d_Y(F, f) \leq \sum_{j \geq 1} \epsilon 2^{-j-1} < \epsilon.$$

For each source point z_0 , choose $U \ni z_0$ and a target chart $\psi : V \rightarrow \mathbb{C}^n$ with $F(\overline{U}) \Subset V$. Uniform convergence gives $f_j(\overline{U}) \subset V$ for large j , and

$$(3.22) \quad \psi \circ f_j \longrightarrow \psi \circ F \quad \text{locally uniformly on } U.$$

The scalar uniform limit theorem applied to (3.22) gives $\psi \circ F \in \mathcal{O}(U, \mathbb{C}^n)$, hence $F \in \mathcal{O}(\mathbb{C}^m, Y)$. $\square$

4. CONVEX APPROXIMATION AND STRONG DOMINABILITY

This section passes from polydiscs to arbitrary compact convex sets and completes the strong dominability criterion. Following the real-face induction of [29, Section 4], we remove the inequalities of an enclosing polytope one at a time. Approximation on each cutting face provides the traces needed for gluing across it.

4.1. Real parameters and approximation on real faces. On a manifold target, finite derivative accuracy is measured in fixed target charts around the compact image of a reference map. Take finitely many compact source pieces K_ν and target charts ψ_ν whose domains contain $H(K_\nu)$ with a positive margin. For a map G with $\psi_\nu \circ G$ defined on K_ν , write

$$\|G - H\|_{C^j(K)} = \max_{\nu} \max_{|\mu| \leq j} \sup_{K_\nu} |D^\mu(\psi_\nu \circ G - \psi_\nu \circ H)|.$$

Derivatives are taken in the displayed real and complex source variables; on a real face they are tangential. Different finite families of charts give equivalent local notions of convergence. For maps jointly holomorphic on a fixed complex neighborhood, uniform approximation on a slightly larger compact neighborhood gives approximation in any fixed finite number of derivatives on the smaller set by Cauchy estimates.

Smooth paths will join the separate slice approximations in Lemma 4.3. The following lemma gives holomorphic approximations on a complex neighborhood of the parameter interval, using Euclidean approximation in a Stein graph neighborhood followed by holomorphic retraction.

Lemma 4.1. *Let $I \subset \mathbb{R}$ be a compact interval and $C \subset \mathbb{C}^k$ a compact convex set, with open neighborhoods $J \subset \mathbb{R}$ and $U \subset \mathbb{C}^k$, respectively. Suppose that*

$$H : J \times U \longrightarrow Y$$

is jointly smooth and that $H_s = H(s, \cdot)$ is holomorphic for every real $s \in J$. For every integer $j \geq 0$ and every $\epsilon > 0$, there exist complex open neighborhoods $V \supset I$, $U_0 \supset C$ and

$$F \in \mathcal{O}(V \times U_0, Y), \quad \|F - H\|_{C^j(I \times C)} < \epsilon.$$

The approximation is also available on prescribed slightly larger products of a compact interval and a compact convex set contained in $J \times U$. The thickness of V may depend on F .

Proof. Choose a compact interval $I^+ = [a, b] \Subset J$, a compact convex set C^+ , and a bounded convex open set X such that

$$(4.1) \quad I \Subset \text{int } I^+, \quad C \Subset \text{int } C^+ \Subset X \Subset U.$$

Apply [20, Corollary 2.2] with parameter set $I^+ \subset \mathbb{R}$, Stein base X , submersion $X \times Y \rightarrow X$, the compact $\mathcal{O}(X)$ -convex set C^+ , and the holomorphic sections $q \mapsto (q, H_s(q))$. This gives a Stein neighborhood Θ of the graph

$$(4.2) \quad \Sigma = \{(s, q, H_s(q)) : s \in I^+, q \in C^+\} \subset \mathbb{C} \times X \times Y.$$

The Remmert–Bishop–Narasimhan embedding theorem and the Docquier–Grauert retraction theorem, in the form used in [9, proof of Lemma 3], give

$$(4.3) \quad e : \Theta \hookrightarrow \mathbb{C}^N, \quad e(\Theta) \text{ closed}, \quad \rho : \Omega \longrightarrow e(\Theta), \quad \rho|_{e(\Theta)} = \text{id},$$

with $\Omega \subset \mathbb{C}^N$ open and ρ holomorphic.

Choose an open set U_1 with $C \Subset U_1 \Subset \text{int } C^+$. Compactness of (4.2) gives

$$(4.4) \quad g(s, q) = e(s, q, H_s(q)), \quad g \in C^\infty(I^+, H^\infty(U_1, \mathbb{C}^N)).$$

Write $\mathcal{B} = H^\infty(U_1, \mathbb{C}^N)$ and $L = j + 1$, so that the derivatives $\partial_s^\ell g$ with $0 \leq \ell < L$ are exactly the s -derivatives of g up to order j . For $x = (s - a)/(b - a)$, form the $\mathcal{B}$ -valued polynomial

$$(4.5) \quad Q_\nu(s) = \sum_{k=0}^{\nu} \partial_s^L g \left(a + \frac{k(b-a)}{\nu} \right) \binom{\nu}{k} x^k (1-x)^{\nu-k}.$$

Let ω be a modulus of continuity of $\partial_s^L g : I^+ \rightarrow \mathcal{B}$ and $M = \sup_{I^+} \|\partial_s^L g\|_{\mathcal{B}}$. The binomial weights in (4.5) have mean x and variance $x(1-x)/\nu$. Splitting the sum at $|a + k(b-a)/\nu - s| = (b-a)\nu^{-1/4}$ therefore gives

$$(4.6) \quad \varepsilon_\nu := \sup_{I^+} \|Q_\nu - \partial_s^L g\|_{\mathcal{B}} \leq \omega((b-a)\nu^{-1/4}) + \frac{M}{2\sqrt{\nu}} \longrightarrow 0.$$

Integrating (4.5) L times, and using the Taylor expansion of P_ν at a to match the first $L - 1$ derivatives of g there, gives a polynomial in s that is holomorphic in q :

$$(4.7) \quad P_\nu(s) = \sum_{\ell=0}^{L-1} \frac{(s-a)^\ell}{\ell!} \partial_s^\ell g(a) + \int_a^s \frac{(s-t)^{L-1}}{(L-1)!} Q_\nu(t) dt.$$

By (4.7), for $0 \leq \ell < L$ the difference $\partial_s^\ell(P_\nu - g)$ is the $(L - \ell)$ -fold primitive of $Q_\nu - \partial_s^L g$ that vanishes to order $L - \ell$ at $s = a$; hence (4.6) implies

$$(4.8) \quad \sup_{I^+} \|\partial_s^\ell(P_\nu - g)\|_{\mathcal{B}} \leq \frac{(b-a)^{L-\ell}}{(L-\ell)!} \varepsilon_\nu.$$

Equation (4.8) and Cauchy estimates across the fixed margin $C \Subset U_1$ yield

$$(4.9) \quad \max_{\ell+|\beta| \leq j} \sup_{I \times C} |\partial_s^\ell \partial_q^\beta (P_\nu - g)| \longrightarrow 0.$$

For large ν , the case $\ell = 0$ of (4.8) places $P_\nu(I \times \overline{U}_0)$ in Ω for some open U_0 with $C \Subset U_0 \Subset U_1$. By compactness of I , there is then a complex neighborhood $V_\nu \supset I$ with $P_\nu(V_\nu \times U_0) \subset \Omega$. Consequently

$$F_\nu = \pi_Y \circ e^{-1} \circ \rho \circ P_\nu \in \mathcal{O}(V_\nu \times U_0, Y).$$

On fixed charts around the compact image, (4.3), (4.4), (4.9), and the chain rule give

$$\|F_\nu - H\|_{C^j(I \times C)} \leq C_j \max_{\ell+|\beta| \leq j} \sup_{I \times C} |\partial_s^\ell \partial_q^\beta (P_\nu - g)| \longrightarrow 0.$$

The margins in (4.1) can be taken around any prescribed compact product inside $J \times U$. $\square$

After complexifying the real parameter, we need approximation that is entire in that variable as well. The reduction of [29, Lemma 4.2] uses a Riemann map to place it in a polydisc factor, then approximates the Riemann map by a polynomial after the entire target-valued approximant has been chosen.

Lemma 4.2. *For an integer $r \geq 0$, suppose that Y satisfies the following property A_r : for every compact convex set $K \subset \mathbb{C}^r$ and every closed polydisc Q , each holomorphic map from a neighborhood of $K \times Q$ to Y can be approximated uniformly on $K \times Q$ by maps entire in all source variables. Then a map holomorphic near $I \times C \times P$, where $I \subset \mathbb{R}$ is a compact interval, $C \subset \mathbb{C}^r$ is compact and convex, and $P \subset \mathbb{C}^q$ is a closed polydisc for an integer $q \geq 0$, can be approximated by maps entire in all displayed complex variables, retaining any fixed finite derivative accuracy on smaller domains.*

Proof. Write the given map as $H(s, y, u)$. Choose a bounded simply connected neighborhood V of I and neighborhoods U_C, U_P such that H is holomorphic on $V \times U_C \times U_P$. Let $\varphi : V \rightarrow \Delta$ be a Riemann map and choose $0 < \rho < 1$ with $\varphi(I) \Subset \Delta_\rho$. The map to which A_r applies is the pullback

$$(4.10) \quad \hat{H}(\zeta, y, u) = H(\varphi^{-1}(\zeta), y, u), \quad (\zeta, y, u) \in \Delta \times U_C \times U_P.$$

In (4.10), treat $\bar{\Delta}_\rho \times P$ as a passive polydisc. After retaining the margins needed for Cauchy estimates, A_r gives an entire $E(\zeta, y, u)$ with

$$(4.11) \quad \|E(\varphi(s), y, u) - H(s, y, u)\|_{C^j(I \times C \times P)} < \epsilon/2.$$

Fix E before the next approximation. Runge's theorem [9, Theorem 2] gives a polynomial p as close to φ as we wish on a slightly larger compact convex neighborhood of I inside V . Then

$$\tilde{H}(s, y, u) = E(p(s), y, u)$$

is entire. On the retained compact set $K = I \times C \times P$, the chain rule in fixed target charts gives

$$(4.12) \quad \begin{aligned} \|\tilde{H} - H\|_{C^j(K)} &\leq \|E(\varphi(s), y, u) - H\|_{C^j(K)} \\ &\quad + C_{E,j} \|p - \varphi\|_{C^j(I)}. \end{aligned}$$

The constant $C_{E,j}$ in (4.12) depends on derivatives of the already fixed E on a compact neighborhood of the two composition images. The first term in (4.12) is less than $\epsilon/2$ by (4.11), and p can be chosen so that the second term is less than $\epsilon/2$. $\square$

The convex slices of a real face vary with its real coordinate. The next construction, from [29, Lemma 4.3], joins their separate approximants into one family on a common domain. Preserving the endpoint maps exactly makes these local paths fit together.

Lemma 4.3. *Assume that A_{r-1} holds, where $r \geq 1$. Let $F \subset \mathbb{R} \times \mathbb{C}^{r-1}$ be compact and convex and P a closed polydisc. Every holomorphic map defined near $F \times P$ can be approximated on this set by entire maps on $\mathbb{C}^r \times \mathbb{C}^{\dim P}$, with control of tangential derivatives of any fixed finite order.*

Proof. Write the variables as (t, y, u) , with t real on the face. Thicken F and P slightly within the original domain of definition. Denote the projection of F onto the t -axis by I and the slices by F_t . For each $t_0 \in I$, choose a compact convex neighborhood C_{t_0} of the whole slice F_{t_0} and a complex disc D_{t_0} about t_0 such that the original map is defined near $D_{t_0} \times C_{t_0} \times P$. The inclusion $F_{t_0} \subset \text{int } C_{t_0}$ persists under perturbation: if not, there would be

$$t_\nu \longrightarrow t_0, \quad (t_\nu, y_\nu) \in F, \quad y_\nu \notin \text{int } C_{t_0}.$$

Compactness gives a subsequence with $(t_\nu, y_\nu) \rightarrow (t_0, y) \in F$, whereas the closedness of $\mathbb{C}^{r-1} \setminus \text{int } C_{t_0}$ gives

$$y \in F_{t_0} \setminus \text{int } C_{t_0} = \emptyset.$$

A finite subdivision of I therefore gives products $D_i \times C_i \times P$ with reserved margins, such that on each subinterval the whole slice F_t lies uniformly inside its assigned C_i . Near each subdivision point, choose

$$F_t \Subset \text{int } C^-, \quad C^- \Subset \text{int } C, \quad C \Subset \text{int}(C_i \cap C_{i+1}),$$

where C^-, C are compact convex intermediate sets with nonempty interior. At a subdivision point, choose a small complex disc D_b about that point, with $D_b \times C \times P$ contained in both adjacent approximation domains and with positive margins. Also choose a common large polydisc B containing all the C_i .

Apply A_{r-1} , treating the t -disc and the u -polydisc as passive parameters, to obtain entire maps g_i that approximate the original map as accurately as we wish on the assigned subinterval. On a slightly larger neighborhood of $D_b \times C \times P$, two adjacent maps g_i, g_{i+1} can be expressed through a chart spray of the original map as

$$g_i = f^\sharp(\cdot, a_i), \quad g_{i+1} = f^\sharp(\cdot, a_{i+1}).$$

Choose a fixed smooth cutoff χ on $[0, 1]$ with $\chi = 0$ near 0 and $\chi = 1$ near 1, and set

$$(4.13) \quad f_\tau = f^\sharp(\cdot, (1 - \chi(\tau))a_i + \chi(\tau)a_{i+1}), \quad 0 \leq \tau \leq 1.$$

The family in (4.13) is defined near C , and equals the entire map g_i or g_{i+1} on the corresponding endpoint subintervals. For $v_\tau = (1 - \chi(\tau))a_i + \chi(\tau)a_{i+1}$,

$$(4.14) \quad \partial_\tau^\ell v_\tau = \chi^{(\ell)}(\tau)(a_{i+1} - a_i) \quad (\ell \geq 1), \quad \|v_\tau\|_{C^j} \leq C_{j,\chi}(\|a_i\|_{C^j} + \|a_{i+1}\|_{C^j}).$$

Since $f^\sharp(\cdot, 0) = f$, the chain rule, (4.14), and uniform bounds for the derivatives of the fixed spray on each retained compact piece K give, for sufficiently accurate choices of g_i and g_{i+1} ,

$$(4.15) \quad \|f_\tau - f\|_{C^j([0,1] \times K)} \leq C_j(\|a_i\|_{C^j(K)} + \|a_{i+1}\|_{C^j(K)}) \longrightarrow 0.$$

We next use rescaling and entire approximation to replace the local path by one available on the common larger polydisc B . Choose $d_* \in \text{int } C$ and

$$(4.16) \quad r_C = \text{dist}(d_*, \partial C) > 0, \quad R_B = \max_{y \in B} |y - d_*|, \quad L > \max(1, R_B/r_C).$$

Equation (4.16) gives

$$d_* + L^{-1}(B - d_*) \Subset \text{int } C.$$

Choose smooth functions $\kappa_0, \kappa_1 : [0, 1] \rightarrow [0, 1]$ with disjoint supports in the intervals where $f_\tau = g_i$ and $f_\tau = g_{i+1}$, respectively, and with $\kappa_0 = 1$ near 0, $\kappa_1 = 1$ near 1. The scaling function is

$$(4.17) \quad \lambda(\tau) = 1 + (L - 1)(\kappa_0(\tau) + \kappa_1(\tau)), \quad 1 \leq \lambda(\tau) \leq L.$$

Set

$$(4.18) \quad \tilde{f}_\tau(t, y, u) = f_\tau(t, d_* + \lambda(\tau)(y - d_*), u).$$

It is well defined near the common small set C : by (4.17), the condition $\lambda > 1$ forces f_τ to be one of the entire endpoint maps. Fix g_i, g_{i+1}, λ, B and chart sprays of both endpoint maps on a neighborhood of $D_b \times B \times P$ before choosing the next approximation.

Apply Lemma 4.1 to $\tilde{f}$ in (4.18), followed by Lemma 4.2 with A_{r-1} , treating the t -disc and u variables as passive. This gives a jointly entire approximation $E(\tau, t, y, u)$. Set

$$(4.19) \quad A_\tau(t, y, u) = E(\tau, t, d_* + \lambda(\tau)^{-1}(y - d_*), u).$$

Since E is entire, (4.19) makes A_τ available on all of B . For $y \in C^-$, set $y_\tau = d_* + \lambda(\tau)^{-1}(y - d_*)$. Then

$$(4.20) \quad y_\tau \in \text{conv}(C^- \cup \{d_*\}) \Subset \text{int } C, \quad \tilde{f}_\tau(t, y_\tau, u) = f_\tau(t, y, u).$$

For a closed disc $\overline{D}'_b \Subset D_b$ about the subdivision point, set $K_- = \overline{D}'_b \times C^- \times P$. Given a derivative order j , choose a compact product K_C in the approximation domain containing all (τ, t, y_τ, u) with $(t, y, u) \in K_-$. By (4.19), (4.20), and the chain rule,

$$(4.21) \quad \|A_\tau - f_\tau\|_{C^j([0,1] \times K_-)} \leq C_j \|E - \tilde{f}\|_{C^j(K_C)},$$

where C_j depends on the fixed scaling function and the retained products. The right side of (4.21) can be made arbitrarily small. Together with (4.15), this makes A_τ arbitrarily close to

the original map f on the retained product. On smaller compact endpoint intervals E_0, E_1 , the scaling satisfies

$$(4.22) \quad \lambda(\tau) = L, \quad S_L(y) = d_* + L^{-1}(y - d_*), \quad S_L(B) \subseteq \text{int } C.$$

Writing $g^{(0)} = g_i$ and $g^{(1)} = g_{i+1}$, equations (4.18) and (4.22) give

$$(4.23) \quad \tilde{f}_\tau(t, S_L(y), u) = g^{(\ell)}(t, y, u), \quad \tau \in E_\ell, \quad y \in B, \quad \ell = 0, 1.$$

Choose K_C also to contain the compact products obtained from the retained endpoint domains by $(\tau, t, y, u) \mapsto (\tau, t, S_L(y), u)$. By (4.19) and (4.23),

$$(4.24) \quad \|A - g^{(\ell)}\|_{C^j(E_\ell \times \overline{D}'_b \times B \times P)} \leq C_{j,L} \|E - \tilde{f}\|_{C^j(K_C)} \longrightarrow 0.$$

Thus (4.24) places A_τ in the previously fixed inverse fiber charts of the endpoint sprays over B .

To concatenate the approximating paths, their endpoint values must agree exactly with g_i and g_{i+1} . On E_ℓ , the inverse fiber coordinates of the fixed endpoint spray give

$$A_\tau = (g^{(\ell)})^\#(\cdot, \beta_\tau^{(\ell)}), \quad \ell = 0, 1.$$

Choose smooth cutoffs $\vartheta_\ell : E_\ell \rightarrow [0, 1]$ equal to 0 near the path endpoint $\tau = \ell$ and to 1 near the other, interior endpoint of E_ℓ . The corrected path is

$$(4.25) \quad \hat{A}_\tau = \begin{cases} (g^{(\ell)})^\#(\cdot, \vartheta_\ell(\tau)\beta_\tau^{(\ell)}), & \tau \in E_\ell, \quad \ell = 0, 1, \\ A_\tau, & \tau \notin E_0 \cup E_1. \end{cases}$$

At the interior endpoint of each E_ℓ , equation (4.25) agrees with A_τ on a neighborhood, so the corrected path is smooth. At the endpoints of the full path interval it satisfies

$$\hat{A}_\tau = g_i \quad \text{for } \tau \text{ near } 0, \quad \hat{A}_\tau = g_{i+1} \quad \text{for } \tau \text{ near } 1.$$

The corrections are available over B by (4.24) and are supported where $\lambda = L$.

Write the subdivision intervals as $I_i = [t_{i-1}, t_i]$. Around each inner division point t_i , choose disjoint joining intervals $J_i = (t_i - \delta_i, t_i + \delta_i) \subseteq D'_b \cap \mathbb{R}$ and fixed smooth functions $\rho_i : J_i \rightarrow [0, 1]$ equal to 0 near the left endpoint and 1 near the right endpoint. Denote the path just constructed between g_i and g_{i+1} by $\hat{A}^{(i)}$. With $I_i^0 = I_i \setminus \bigcup_\ell J_\ell$, the family on the common domain is

$$(4.26) \quad H(t, y, u) = \begin{cases} \hat{A}_{\rho_i(t)}^{(i)}(t, y, u), & t \in J_i, \\ g_i(t, y, u), & t \in I_i^0. \end{cases}$$

Near the ends of J_i , (4.25) makes the first expression in (4.26) equal to g_i and g_{i+1} , respectively. Thus H is smooth in t and holomorphic in (y, u) near $I \times B \times P$. For the compact pieces

$$\mathcal{K}_i = (F \cap (I_i \times \mathbb{C}^{r-1})) \times P, \quad \mathcal{J}_i = (F \cap (\overline{J}_i \times \mathbb{C}^{r-1})) \times P,$$

and any prescribed derivative order j , the chain rule and $\rho_i(J_i) \subset [0, 1]$ give

$$(4.27) \quad \begin{aligned} \|H - f\|_{C^j(F \times P)} &\leq C_j \max_i \|g_i - f\|_{C^j(\mathcal{K}_i)} \\ &\quad + C_j \max_i \|\hat{A}^{(i)} - f\|_{C^j([0,1] \times \mathcal{J}_i)}, \end{aligned}$$

where f is independent of the path parameter and C_j depends on the fixed ρ_i . The two errors in (4.27) can be made arbitrarily small, the first by the local entire approximation and the second by (4.15), (4.21), (4.24), and (4.25). Extend H slightly beyond the endpoints of I by its first and last entire maps. Apply Lemma 4.1 again to obtain H_1 , jointly holomorphic near a complex neighborhood of $I \times B \times P$. Since B is a polydisc, Lemma 4.2 applied to H_1 , with A_{r-1} and the u -variables as passive parameters, now gives an entire map in all variables. When $r = 1$, there is no y -variable and the rescaling step is omitted. $\square$

4.2. Real-interface gluing and convex approximation. Fix $0 < \alpha < 1$. For a subset M of a Euclidean space and a vector-valued function u , write

$$(4.28) \quad \|u\|_{C^\alpha(M)} = \sup_M |u| + [u]_{\alpha, M}, \quad [u]_{\alpha, M} = \sup_{\substack{z, w \in M \\ z \neq w}} \frac{|u(z) - u(w)|}{|z - w|^\alpha}.$$

For $Q \subset \mathbb{C}^m$ compact and convex, write

$$(4.29) \quad Q_r = \{z \in \mathbb{C}^m : \text{dist}(z, Q) < r\}, \quad r > 0.$$

The interface and its two sides are

$$(4.30) \quad M_r = Q_r \cap \{\text{Im } z_1 = 0\}, \quad Q_r^\pm = Q_r \cap \{\pm \text{Im } z_1 > 0\}.$$

The norms used for traces and for the two sides are

$$(4.31) \quad \begin{aligned} \|u\|_{r, b, \alpha} &= \sup_{v \in B_b} \|u(\cdot, v)\|_{C^\alpha(M_r)}, \\ \|a\|_{r, b, *}^\pm &= \sup_{Q_r^\pm \times B_b} |a| + \sup_{v \in B_b} \|a(\cdot, v)\|_{M_r} \|_{C^\alpha(M_r)}. \end{aligned}$$

The difference quotient in (4.28) always uses all base variables. The fiber variable v is subject to a supremum and is treated by holomorphic estimates on smaller balls.

The remaining task is to turn close traces on the cutting face into maps that agree exactly there. The additive splitting of [29, Lemma 4.4] plays the role of Lemma 2.5 in the full-base C^α norms (4.31). Its nonlinear correction gives the following real-interface gluing theorem [29, Lemma 4.5], the spray form of [29, Lemma 3.2]. The corrections also control the change on the old side, allowing successive cuts with a prescribed total error.

Lemma 4.4. *For a compact convex $Q \subset \mathbb{C}^m$ with the geometry (4.29)–(4.30), fix $r, b > 0$, an outer fiber margin $\mu > 0$, losses $0 < \sigma < r/2$, $0 < \tau < b$, and $\eta > 0$. There exists $\delta \in (0, \mu)$ such that the following holds. Suppose that F_-, F_+ are holomorphic on $Q_r^- \times B_{b+\mu}$, $Q_r^+ \times B_b$, continuous up to their interface, and their traces satisfy on $M_r \times B_b$*

$$F_-(z, \gamma(z, v)) = F_+(z, v), \quad \gamma(z, v) = v + c(z, v), \quad \|c\|_{r, b, \alpha} < \delta.$$

The transition is holomorphic in the complex tangential variables and in v . Under these hypotheses there are holomorphic fiber reparametrizations $\alpha_\pm = \text{id} + u_\pm$ on $Q_{r-\sigma}^\pm \times B_{b-\tau}$, with C^α traces and values in B_b , such that

$$\begin{aligned} \|u_-\|_{r-\sigma, b-\tau, *}^- + \|u_+\|_{r-\sigma, b-\tau, *}^+ &< \eta, \\ F_-(z, \alpha_-(z, v)) &= F_+(z, \alpha_+(z, v)) \quad (z \in M_{r-\sigma}). \end{aligned}$$

The two corrected maps join holomorphically across the interface. If F is their core and $K \Subset Q_{r-\sigma}^- \cup M_{r-\sigma}$, then

$$(4.32) \quad \sup_K d_Y(F, F_-(\cdot, 0)) \leq \omega_{-, K}(\eta),$$

where $\omega_{-, K}$ is a continuity modulus of the fixed spray F_- in its fiber variable.

Proof of Theorem 2.2. Recall that A_r allows a compact convex factor in $\mathbb{C}^r$ together with arbitrary polydisc factors. We prove it by induction on r . The base case A_0 follows from Proposition 3.2. Assume that A_{r-1} holds.

First consider extension across a single real affine hyperplane. For a nonconstant real affine function,

$$(4.33) \quad \ell(z) = \text{Re} \sum_{j=1}^r a_j z_j + c, \quad w_1 = i \left(\sum_{j=1}^r a_j z_j + c - b \right), \quad \text{Im } w_1 = \ell(z) - b.$$

Complete the nonzero linear part of w_1 to a basis of complex linear functionals and retain the translation in (4.33). After renaming the resulting affine coordinates,

$$Q_- = Q \cap \{\operatorname{Im} z_1 \leq 0\} \subset Q \subset \mathbb{C}^r,$$

where Q is compact and convex and f is holomorphic near $Q_- \times P$. When Q_- has nonempty interior and $Q_- \neq Q$, the cutting hyperplane meets $\operatorname{int} Q$; this is the case treated here. Take a slightly larger passive polydisc $P^+ \supseteq P$. For small $r_* > 0$,

$$(4.34) \quad (\overline{Q_{r_*}} \cap \{\operatorname{Im} z_1 \leq 0\}) \times P^+ \subset \operatorname{dom} f.$$

Indeed, if (4.34) failed for every $r_* > 0$, there would be $r_\nu \downarrow 0$ and

$$z_\nu \in \overline{Q_{r_\nu}}, \quad \operatorname{Im}(z_\nu)_1 \leq 0, \quad u_\nu \in P^+, \quad (z_\nu, u_\nu) \notin \operatorname{dom} f.$$

A subsequence converges to $(z, u) \in Q_- \times P^+$. Since this compact product lies inside the open domain of f , eventually $(z_\nu, u_\nu) \in \operatorname{dom} f$, a contradiction.

Lemma 2.4, applied on the convex lower-side domain supplied by (4.34), gives a fixed chart spray $f^\sharp(z, u, v)$. By Lemma 4.3, with u and v as passive parameters, its trace on a slightly larger real face can be approximated by an entire map $G(z, u, v)$ with control of tangential derivatives through order two. On the interface, invert the fixed fiber charts of $f^\sharp$ to obtain a transition $\gamma = \operatorname{id} + c$; the fixed inverse charts and the chain rule give, on a retained face,

$$(4.35) \quad \|c\|_{C^1} \leq C \|G - f^\sharp\|_{C^1} \longrightarrow 0.$$

The norms in (4.35) are measured in the fixed charts and include all base variables (z, u) , while the derivatives in u follow from Cauchy estimates on the reserved parameter margin. Since the face is convex and has bounded diameter D_0 ,

$$(4.36) \quad \|c\|_{C^\alpha} \leq \|c\|_{C^0} + D_0^{1-\alpha} \|D_{z,u} c\|_{C^0} \longrightarrow 0.$$

For the gluing, the compact convex base and its complex dimension are

$$(4.37) \quad \mathcal{Q} = Q \times P, \quad m = r + \dim P,$$

with coordinates (z, u) ; only $v \in \mathbb{C}^n$ remains a fiber variable. Small Euclidean thickenings of $\mathcal{Q}$ lie in the reserved product neighborhoods, and the C^α norm of the transition includes differences in all (z, u) variables. By (4.35), (4.36), and the Cauchy estimates in u ,

$$(4.38) \quad \sup_{v \in \overline{B_b}} \|c(\cdot, v)\|_{C^\alpha(\mathcal{Q}_{r_*} \cap \{\operatorname{Im} z_1 = 0\})} \longrightarrow 0.$$

Thus Lemma 4.4 applies on $\mathcal{Q}_{r_*}$, with the base (4.37) and any prescribed loss $0 < \sigma < r_*/2$, by (4.38). The resulting corrections satisfy

$$(4.39) \quad \gamma(z, u, \alpha_+(z, u, v)) = \alpha_-(z, u, v) \quad \text{on the interface}, \quad \|\alpha_- - \operatorname{id}\|_* + \|\alpha_+ - \operatorname{id}\|_* < \eta.$$

The resulting map on $\mathcal{Q}_{r_*-\sigma}$ is

$$(4.40) \quad \tilde{f}(z, u) = \begin{cases} f^\sharp(z, u, \alpha_-(z, u, 0)), & \operatorname{Im} z_1 \leq 0, \\ G(z, u, \alpha_+(z, u, 0)), & \operatorname{Im} z_1 \geq 0. \end{cases}$$

The two traces in (4.40) agree by (4.39), so Lemma 4.4 makes $\tilde{f}$ holomorphic near $Q \times P$. Equation (4.32) gives

$$(4.41) \quad \sup_{Q_- \times P} d_Y(\tilde{f}, f) \leq \omega_{f^\sharp, Q_- \times P}(\eta) \longrightarrow 0 \quad (\eta \longrightarrow 0).$$

To obtain approximation on an arbitrary compact convex set $K \subset \mathbb{C}^r$, take a map f holomorphic near $K \times P$. After thickening K and P inside the original domain, choose a full-dimensional real convex polytope K' with

$$K \Subset \operatorname{int} K', \quad K' \times P \Subset \operatorname{dom} f.$$

Choose R with $K' \Subset \text{int } \overline{\Delta}_R^r$ and real affine inequalities $\ell_j(z) \leq b_j$ such that

$$(4.42) \quad K' = \bigcap_{j=1}^M \{\ell_j \leq b_j\}, \quad Q_k = \overline{\Delta}_R^r \cap \bigcap_{j=k+1}^M \{\ell_j \leq b_j\} \quad (0 \leq k \leq M),$$

so that

$$(4.43) \quad Q_0 = K', \quad Q_M = \overline{\Delta}_R^r, \quad Q_{k-1} = Q_k \cap \{\ell_k \leq b_k\}.$$

Every Q_k in (4.42) is compact and convex and contains the full-dimensional set K' , and whenever a step enlarges the set, its cutting hyperplane meets $\text{int } Q_k$. By (4.33), the cut in (4.43) reduces to the case treated above. Applying (4.41) successively, starting with $f_0 = f$, gives f_k holomorphic near $Q_k \times P$ with

$$(4.44) \quad \sup_{Q_{k-1} \times P} d_Y(f_k, f_{k-1}) < \epsilon_k, \quad \sum_{k=1}^M \epsilon_k < \epsilon/2.$$

Using (4.44), Proposition 3.2 gives an entire F with

$$\sup_{\overline{\Delta}_R^r \times P} d_Y(F, f_M) < \epsilon/2, \quad \sup_{K \times P} d_Y(F, f) < \epsilon.$$

This proves A_r . Setting $q = 0$ in the statement, so that no passive parameters remain, gives CAP in every source dimension. $\square$

Proof of Theorem 1.3. Suppose first that Y is strongly dominable. Lemma 2.3, Theorem 2.2 and Forstnerič's characterization [12, Theorem 0.1] give

$$\text{strong dominability} \implies \mathcal{L}_n \implies \text{CAP} \implies \text{Oka}.$$

Conversely, let Y be Oka and $y \in Y$. Choose a coordinate parametrization $\phi : B_R \rightarrow Y$ with $\phi(0) = y$ and $d\phi_0$ invertible. For $0 < r < R$ and a continuous cutoff $\chi : [0, \infty) \rightarrow [0, 1]$ equal to 1 on $[0, r]$ and 0 on $[R, \infty)$, the map

$$f_0(z) = \begin{cases} \phi(\chi(|z|)z), & |z| < R, \\ y, & |z| \geq R \end{cases}$$

is continuous on $\mathbb{C}^n$ and agrees with ϕ on B_r . The Oka property with first-order jet interpolation at 0 [11, Corollary 1.3] gives

$$(4.45) \quad H \in \mathcal{O}(\mathbb{C}^n, Y), \quad j_0^1 H = j_0^1 \phi, \quad H(0) = y, \quad dH_0 = d\phi_0.$$

Equation (4.45) shows that Y is dominable at every y . $\square$

Forstnerič–Lárusson's stratification theorem supplies the pointwise domination needed for the following consequence.

Corollary 4.5. *Every complex manifold with a finite Oka stratification is Oka.*

Proof. For the given finite stratification, [18, Theorem 2] gives

$$\forall x \in X \exists H_x \in \mathcal{O}(\mathbb{C}^{\dim X}, X) : \quad H_x(0) = x, \quad d(H_x)_0 \text{ is invertible.}$$

Thus X is strongly dominable, and Theorem 1.3 gives the Oka property. $\square$

5. EXTENSION ACROSS A PROPER ANALYTIC SUBSET

We prove the union theorem by propagating $\mathcal{L}_n$ across a proper analytic subset. The argument uses holomorphic families on scalar discs of increasing radius, with convergence required only near scalar zero.

For a complex manifold Y of dimension $n \geq 1$, let

$$\mathcal{G} = \{x \in Y : Y \text{ has property } \mathcal{L}_n \text{ at } x\}.$$

The set $\mathcal{G}$ is open: every point of a witness neighborhood has that same witness neighborhood. We will show that $Y \setminus \mathcal{G}$ is empty if it is locally contained in a proper analytic subset.

In this section the fiber domains are the maximum-norm polydiscs

$$\Delta_r^n = \{w \in \mathbb{C}^n : \|w\|_\infty < r\};$$

vector norms are the maximum norm and matrix norms its induced operator norm. Closed polydiscs retain the notation $\overline{\Delta}_r^n$. All maps near compact products are restricted, when needed, to a product of an open planar neighborhood and an open polydisc. These sources satisfy the triviality hypothesis of Lemma 2.4. Its fiber-preserving coordinates and uniform inverse-chart margins will be used below.

The next lemma adapts the scalar-substitution argument of An–Xie [3, Lemma 4.3]. Substitution of a holomorphic scalar function preserves approximability on the original source set. The increasing radii accommodate polynomial approximants that may be large away from that set.

Lemma 5.1. *Fix a special planar disc pair $D \subset D'$ and a closed polydisc $P_* \subset \mathbb{C}^q$, where $q \geq 0$. Write*

$$K = D \times P_*, \quad L = D' \times P_*,$$

write $s = (z, p)$ for a source point, and let $\mathcal{A}$ consist of the maps in $\mathcal{O}(\text{Op } K, Y)$ uniformly approximable on K by maps in $\mathcal{O}(\text{Op } L, Y)$. Suppose $f_0 \in \mathcal{A}$, Ω is an open neighborhood of K , and

$$G : \Omega \times \Delta_r^n \rightarrow Y, \quad G(s, 0) = f_0(s),$$

is holomorphic, with $(s, w) \mapsto (s, G(s, w))$ biholomorphic onto its image. Suppose

$$T_\nu : \Omega \times \Delta_r^n \times \Delta_{R_\nu} \rightarrow Y, \quad T_\nu(s, w, 0) = G(s, w), \quad R_\nu \rightarrow \infty,$$

are holomorphic and converge locally uniformly on $\Omega \times \Delta_r^n \times \Delta_\tau$, for some $\tau > 0$, to a holomorphic map T . For every $\varphi \in \mathcal{O}(\text{Op } K, \mathbb{C})$ with $\|\varphi\|_K < \tau$, the map

$$s \mapsto T(s, 0, \varphi(s))$$

belongs to $\mathcal{A}$.

Proof. Fix $\epsilon > 0$ and numbers $\|\varphi\|_K < t_0 < t_1 < \tau$. Since $f_0 \in \mathcal{A}$, there is $g \in \mathcal{O}(\text{Op } L, Y)$ so close to f_0 on K that

$$\rho(s) = G(s, \cdot)^{-1}(g(s))$$

is holomorphic near K , $\|\rho\|_K < r/16$, and

$$(5.1) \quad \sup_K d_Y(T(s, \rho(s), \varphi(s)), T(s, 0, \varphi(s))) < \epsilon/4.$$

The last assertion follows from uniform continuity on a compact neighborhood of $K \times \{0\} \times \varphi(K)$. Lemma 2.4 supplies a family $F(s, v)$ centered at g near L , available for $v \in \Delta_{2b}^n$, with inverse fiber coordinates Ψ_s .

We specify the planar geometry used for the two families. Write $D' = D \cup E$, where E is the attached disc. In E , a crosscut close to $D \cap E$, but disjoint from D , separates a small collar of D from the rest of E . A narrow closed neighborhood of that crosscut and small thickenings of the two sides give compact Jordan regions A, B such that

$$D \subset \text{int } B, \quad D' \subset \text{int}(A \cup B), \quad C = A \cap B \text{ is a closed Jordan disc disjoint from } D,$$

$$\overline{A \setminus B} \cap \overline{B \setminus A} = \emptyset.$$

The collar can be as thin as needed, so $B \times P_*$ lies in the common domain of G, ρ, φ , and F is available near $A \times P_*$. There are disjoint closed Jordan neighborhoods D^+, C^+ of D, C , a compact neighborhood B^+ of B , and polydiscs

$$P_* \Subset \text{int } P^+ \Subset \text{int } P^{++}$$

such that

$$D \Subset \text{int } D^+, \quad C \Subset \text{int } C^+, \quad D^+ \cup C^+ \subset B^+, \quad B \Subset \text{int } B^+,$$

and $B^+ \times P^{++}$ stays in those common domains, with $\|\rho\| < r/8$ there. The family F is available near both $A \times P^{++}$ and $C^+ \times P^{++}$. All these choices precede the index ν and all polynomial approximations below.

Choose fixed open planar neighborhoods U_A, U_B of A, B that cover a neighborhood of D' and have separated fringes. They can be taken with

$$U_A \cap U_B \Subset \text{int } C^+, \quad U_B \Subset \text{int } B^+,$$

and within the relevant family domains. A fixed open polydisc U_P satisfies $P_* \Subset U_P \Subset \text{int } P^+$. Lemma 2.5 applies with a fixed constant C_0 .

The sets $C^+ \times P^{++}$ and $(D^+ \cup C^+) \times P^+$ are polynomially convex. Indeed, the planar factors have connected complements and polynomial hulls commute with products. This will allow a matrix approximation on the overlap and a scalar approximation with different data near D and C .

On a neighborhood of $(s, \rho(s), 0)$ over $C^+ \times P^{++}$, consider

$$\mathcal{H}_\nu(s, w, u) = \Psi_s(T_\nu(s, w, u)).$$

The exact normalization at scalar zero gives

$$(5.2) \quad \mathcal{H}_\nu(s, \rho(s), 0) = 0, \quad E(s) := \partial_w \mathcal{H}_\nu(s, \rho(s), 0) = \partial_w(\Psi_s \circ G)(s, \rho(s)) \in \text{GL}_n(\mathbb{C}).$$

In particular, both the center and E are independent of ν . The inverse chart of F contains a fixed neighborhood of the compact graph of g on this product. Since $T(s, \rho(s), 0) = g(s)$, continuity first gives a common smaller neighborhood in (w, u) for the limit. Local uniform convergence then places all sufficiently large T_ν in the same inverse chart there. On still smaller compact products, Cauchy estimates give uniform bounds for the first and second derivatives of $\mathcal{H}_\nu$. Likewise, on a fixed compact neighborhood of

$$K \times \overline{\Delta_{r/2}^n} \times \overline{\Delta_{t_1}},$$

the first derivatives of T_ν in w, u are uniformly bounded in fixed finitely many target charts containing compact pieces of the limit image with positive margins. The source compact sets and target charts are independent of ν , and the same charts contain the corresponding images of all sufficiently large T_ν .

Oka–Weil approximation [9, Theorem 18] and Taylor truncation give a polynomial matrix L_B with

$$\|L_B - E^{-1}\|_{C^+ \times P^{++}} < \eta, \quad 0 < \eta \leq 1.$$

The two useful bounds are

$$\|L_B\|_{C^+ \times P^{++}} \leq N := 1 + \|E^{-1}\|_{C^+ \times P^{++}}, \quad M := \max\{1, \|L_B\|_{B^+ \times P^+}\} < \infty.$$

Here N is independent of the chosen approximating polynomial, whereas M is fixed only after that polynomial has been chosen.

The derivative bounds on the fixed target charts and uniform continuity of their inverse coordinate maps give moduli ω_w, ω_u , independent of all sufficiently large ν , such that

$$(5.3) \quad \begin{aligned} d_Y(T_\nu(s, w, u), T_\nu(s, w', u)) &\leq \omega_w(\|w - w'\|_\infty), \\ d_Y(T_\nu(s, w, u), T_\nu(s, w, u')) &\leq \omega_u(|u - u'|) \end{aligned}$$

on $K \times \overline{\Delta_{r/2}^n} \times \overline{\Delta_{t_1}}$, with $\omega_w(h), \omega_u(h) \rightarrow 0$ as $h \rightarrow 0$. Choose $\delta > 0$ sufficiently small that the uniform overlap Taylor estimates apply and

$$(5.4) \quad \delta < b/2, \quad \delta M < r/8, \quad \omega_w(\delta M/4) < \epsilon/4.$$

Only after δ is chosen, take a scalar polynomial u_B with

$$(5.5) \quad \|u_B - \varphi\|_{D^+ \times P^+} < e_1, \quad \|u_B\|_{C^+ \times P^+} < e_0.$$

This follows from the Oka–Weil theorem [9, Theorem 18] on the disjoint union, since the data φ and 0 are holomorphic on disjoint neighborhoods. Choose e_1 so that

$$(5.6) \quad e_1 < t_0 - \|\varphi\|_K, \quad \omega_u(e_1) < \epsilon/4.$$

Then (5.5) and (5.6) give $|u_B| < t_0$ on K . The number e_0 will be chosen relative to δ .

On the two sides consider the families

$$F_A(s, \zeta) = F(s, \delta\zeta), \quad F_{B,\nu}(s, \zeta) = T_\nu(s, \rho(s) + \delta L_B(s)\zeta, u_B(s)).$$

The first is available for $\zeta \in \Delta_4^n$. On the second side,

$$\|\rho(s) + \delta L_B(s)\zeta\| < r/8 + 2\delta M < r/2 \quad (\zeta \in \Delta_2^n).$$

On the overlap its transition is

$$\gamma_\nu(s, \zeta) = \delta^{-1} \Psi_s(F_{B,\nu}(s, \zeta)).$$

Taylor expansion at $(w, u) = (\rho(s), 0)$, using (5.2), gives

$$(5.7) \quad \|\gamma_\nu - \text{id}\|_{(U_A \cap U_B) \times U_P \times \Delta_2^n} \leq c(\eta + \delta + e_0/\delta),$$

uniformly for large ν . Indeed, (5.2) gives the exact expansion

$$(5.8) \quad \begin{aligned} \mathcal{H}_\nu(s, \rho(s) + \delta L_B(s)\zeta, u_B(s)) &= \delta E(s) L_B(s)\zeta + \mathcal{R}_\nu(s, \zeta), \\ \|\mathcal{R}_\nu\|_\infty &\leq C(\delta^2 N^2 + e_0), \quad \|EL_B - I\|_\infty \leq \|E\|_\infty \eta. \end{aligned}$$

The quadratic term in (5.8) uses the uniform second derivative bound in w , and the term Ce_0 uses the first derivative bound in u and (5.5). Dividing (5.8) by δ and using $\|\zeta\|_\infty < 2$ gives (5.7), with c depending on the fixed products, N , and the uniform derivative bounds.

Choose η , then L_B and M , then δ , and finally e_0 so that (5.7) is small enough for the planar gluing construction of Section 2, with the smallness conditions (2.7) on the smaller fiber polydisc Δ_1^n . Cauchy estimates from Δ_2^n control the derivative in (2.7). The corrections can be required to have norms less than $1/4$. All previous upper bounds on δ are compatible with this order of choices.

The increasing scalar radii enter here: the polynomial u_B is already fixed and has a finite norm on $B^+ \times P^+$. Choose ν last, so large that

$$R_\nu > 1 + \|u_B\|_{B^+ \times P^+}$$

and

$$(5.9) \quad \sup_K d_Y(T_\nu(s, \rho(s), \varphi(s)), T(s, \rho(s), \varphi(s))) < \epsilon/4.$$

Thus $u_B(B^+ \times P^+) \subset \Delta_{R_\nu}$, so $F_{B,\nu}$ is available on the full fixed B side. On the overlap, (5.5) keeps u_B in the fixed small scalar disc on which (5.8) holds.

The construction (2.6)–(2.10) supplies a, b on the two sides with

$$F_A(s, a(s)) = F_{B,\nu}(s, b(s)), \quad \|a\|, \|b\| < 1/4.$$

These expressions give a map $\tilde{g}$ holomorphic near L . On $K \subset B \times P_*$ it is

$$\tilde{g}(s) = T_\nu(s, \rho(s) + \delta L_B(s)b(s), u_B(s)).$$

For $s \in K$, the triangle inequality gives

$$\begin{aligned}
& d_Y(\tilde{g}(s), T(s, 0, \varphi(s))) \\
& \leq d_Y(\tilde{g}(s), T_\nu(s, \rho(s), u_B(s))) \\
& \quad + d_Y(T_\nu(s, \rho(s), u_B(s)), T_\nu(s, \rho(s), \varphi(s))) \\
& \quad + d_Y(T_\nu(s, \rho(s), \varphi(s)), T(s, \rho(s), \varphi(s))) \\
& \quad + d_Y(T(s, \rho(s), \varphi(s)), T(s, 0, \varphi(s))) \\
& \leq \omega_w(\delta M/4) + \omega_u(e_1) + \epsilon/4 + \epsilon/4 < \epsilon.
\end{aligned}$$

The four bounds follow from (5.3), (5.4), (5.6), (5.9), and (5.1). Since $\tilde{g} \in \mathcal{O}(\text{Op}(D' \times P_*), Y)$, the map $T(\cdot, 0, \varphi)$ belongs to $\mathcal{A}$. $\square$

Scalar substitution can now be iterated in independent tangent directions. Composing the limiting families gives local coordinates at x , so a map with image in a sufficiently small neighborhood of x can be obtained from the constant map by successive substitutions.

Proposition 5.2. *Let $x \in Y$, $\dim Y = n \geq 1$. Suppose on one neighborhood $V \ni x$ there are holomorphic maps*

$$h_{j,\nu} : V \times \Delta_{R_\nu} \rightarrow Y, \quad h_{j,\nu}(y, 0) = y, \quad 1 \leq j \leq n, \quad R_\nu \rightarrow \infty.$$

Suppose these maps converge locally uniformly on $V \times \Delta_\tau$, for some $\tau > 0$, to holomorphic maps h_j , and that $\partial_t h_j(x, 0)$, $1 \leq j \leq n$, form a basis of $T_x Y$. Then $x \in \mathcal{G}$.

Proof. The limits satisfy $h_j(y, 0) = y$. For $a = (a_1, \dots, a_n)$ near zero, consider the exact compositions

$$\theta_0(a) = x, \quad \theta_j(a) = h_j(\theta_{j-1}(a), a_j), \quad \Theta(a) = \theta_n(a).$$

On a sufficiently small polydisc $Q \ni 0$, every intermediate value belongs to a fixed relatively compact subset of V , and $|a_j| < \tau$. Since $d_y h_j(y, 0) = \text{id}$, the columns of $d\Theta_0$ are $\partial_t h_j(x, 0)$. The inverse function theorem gives a fixed neighborhood $W \ni x$ on which Θ^{-1} is available with image relatively compact in Q .

For arbitrary data $f \in \mathcal{O}(\text{Op}(D \times P), W)$ in the definition of $\mathcal{L}_n$, choose

$$P_0 \Subset \text{int } P_* \Subset \text{int } P, \quad (a_1, \dots, a_n) = \Theta^{-1} \circ f,$$

and consider near $K = D \times P_*$ the maps

$$f_0 = x, \quad f_j = h_j(f_{j-1}, a_j), \quad 1 \leq j \leq n.$$

They are exact holomorphic maps, $f_n = f$, and all their values lie in V ; also $\|a_j\|_K < \tau$. The constant map f_0 is in the approximation class $\mathcal{A}$ of Lemma 5.1.

If $f_{j-1} \in \mathcal{A}$, Lemma 2.4 gives coordinates $G_j(s, w)$ centered at f_{j-1} , with their whole image in V after restriction of the fixed base neighborhood and the fiber radius. The families

$$T_{j,\nu}(s, w, t) = h_{j,\nu}(G_j(s, w), t), \quad T_j(s, w, t) = h_j(G_j(s, w), t)$$

satisfy every hypothesis of Lemma 5.1; in particular their scalar-zero value is exactly $G_j(s, w)$. With $\varphi = a_j$, that lemma yields $f_j \in \mathcal{A}$. Finite induction gives $f \in \mathcal{A}$, and restriction to P_0 proves $\mathcal{L}_n$ at x . $\square$

We obtain the required families from a coordinate disc whose boundary lies in $\mathcal{G}$. Planar extension enlarges the scalar domain; inverse branches then recenter the approximants so that scalar zero gives the identity exactly.

Proposition 5.3. *Let $\vartheta : U \rightarrow \Omega \subset \mathbb{C}^n$ be a coordinate chart, $x \in U$, and $v \neq 0$. Suppose for some $r > 0$ that*

$$\{\vartheta(x) + \zeta v : |\zeta| \leq r\} \subset \Omega, \quad \vartheta^{-1}(\{\vartheta(x) + \zeta v : |\zeta| = r\}) \subset \mathcal{G}.$$

Then $x \in \mathcal{G}$.

Proof. The closed disc is compactly contained in the chart, and its boundary is a compact subset of the open set $\mathcal{G}$. Hence all sufficiently small changes of the direction v preserve both properties. Choose n such directions $v_1, \dots, v_n$ which are linearly independent. Their choice and nonzero determinant are fixed before any approximation.

For each j , complete v_j to a basis $(v_j, e_{j,2}, \dots, e_{j,n})$ of $\mathbb{C}^n$ and consider

$$\psi_j(z, p) = \vartheta^{-1} \left(\vartheta(x) + z v_j + \sum_{k=2}^n p_{k-1} e_{j,k} \right).$$

A sufficiently small closed polydisc $P_j \subset \mathbb{C}^{n-1}$ makes this a coordinate parametrization near $\overline{\Delta}_r^1 \times P_j$. Moreover, for each $\zeta \in \partial \Delta_r$, the whole fiber $\psi_j(\{\zeta\} \times P_j)$ lies in a coordinate witness neighborhood for $\mathcal{L}_n$. To obtain one P_j for all ζ , first cover the boundary of the central disc by finitely many witness neighborhoods, use continuity to obtain corresponding source arcs, and then shrink the transverse polydisc so that the compact arc products stay in those neighborhoods.

Choose $P'_j \Subset \text{int } P_j$ centered at zero and a sequence $R_\nu > r$ with $R_\nu \rightarrow \infty$. Proposition 2.6, with errors tending to zero, gives

$$\sigma_{j,\nu} \in \mathcal{O}(\text{Op}(\overline{\Delta}_{R_\nu}^1 \times P'_j), Y), \quad \sigma_{j,\nu} \longrightarrow \psi_j \quad \text{uniformly on } \overline{\Delta}_r^1 \times P'_j.$$

On a smaller fixed product around $(0, 0)$, the images lie in the target chart, and Cauchy estimates give convergence of the first derivatives. Choose $h_j > 0$ with $\overline{\Delta}_{h_j}^n \Subset \Delta_r \times \text{int } P'_j$. In the coordinates of ψ_j , write

$$(5.10) \quad A_{j,\nu} = \psi_j^{-1} \circ \sigma_{j,\nu} = \text{id} + e_{j,\nu}, \quad \|e_{j,\nu}\|_{\overline{\Delta}_{h_j}^n} < h_j/4, \quad \|De_{j,\nu}\|_{\overline{\Delta}_{h_j}^n} < 1/2$$

for all sufficiently large ν . For $b \in \Delta_{h_j/2}^n$, the map $a \mapsto b - e_{j,\nu}(a)$ satisfies, by (5.10),

$$\|b - e_{j,\nu}(a)\|_\infty < 3h_j/4, \quad \|e_{j,\nu}(a) - e_{j,\nu}(a')\|_\infty \leq \frac{1}{2} \|a - a'\|_\infty.$$

It is therefore a contraction of $\overline{\Delta}_{h_j}^n$ into its interior. Its unique fixed point $a_{j,\nu}(b)$ satisfies

$$A_{j,\nu}(a_{j,\nu}(b)) = b, \quad \|a_{j,\nu}(b) - b\|_\infty \leq \|e_{j,\nu}\|_{\overline{\Delta}_{h_j}^n} \longrightarrow 0.$$

The invertibility of $DA_{j,\nu}$ in (5.10) makes $a_{j,\nu}$ holomorphic. On the fixed neighborhood

$$V = \bigcap_{j=1}^n \psi_j(\Delta_{h_j/2}^n)$$

of x , this gives inverse branches

$$\ell_{j,\nu} = a_{j,\nu} \circ \psi_j^{-1} = (s_{j,\nu}, p_{j,\nu}), \quad \sigma_{j,\nu} \circ \ell_{j,\nu} = \text{id}_V.$$

After another fixed restriction of V , the images of all these inverse branches lie in compact subproducts of $\Delta_r \times \text{int } P'_j$. They converge to ψ_j^{-1} , and there is $C > 0$ such that $|s_{j,\nu}(y)| < C$ for every $y \in V$ and every retained index. The maps

$$h_{j,\nu}(y, t) = \sigma_{j,\nu}(s_{j,\nu}(y) + t, p_{j,\nu}(y))$$

are consequently holomorphic on $V \times \Delta_{R_\nu - C}$ and satisfy

$$h_{j,\nu}(y, 0) = y$$

exactly. The radii $R_\nu - C$ tend to infinity.

There is one $\tau > 0$ such that, for $y \in V$ and $|t| < \tau$, all translated source points remain in a fixed compact subproduct of the original convergence product. Thus these maps converge locally uniformly on $V \times \Delta_\tau$ to

$$h_j(y, t) = \psi_j(\psi_j^{-1}(y) + (t, 0)).$$

Their derivatives $\partial_t h_j(x, 0)$ are $(d\vartheta_x)^{-1}v_j$ and form a basis. Proposition 5.2 gives $x \in \mathcal{G}$. $\square$

A suitable coordinate line through a point of a proper analytic subset avoids the subset on a small punctured disc, providing the boundary condition in Proposition 5.3.

Corollary 5.4. *Suppose every point of Y has a connected neighborhood on which $Y \setminus \mathcal{G}$ is contained in a proper complex analytic subset. Then $\mathcal{G} = Y$.*

Proof. Consider a point $x \notin \mathcal{G}$. In a connected coordinate neighborhood centered at x , let B be a proper analytic subset containing the local complement of $\mathcal{G}$. Its ideal has a nonzero holomorphic germ a at x . In centered coordinates, write its first nonzero homogeneous term as a_m . Choose $v \in \mathbb{C}^n$ with $a_m(v) \neq 0$. Then

$$a(\zeta v) = \zeta^m a_m(v) + O(\zeta^{m+1})$$

is not identically zero. On a sufficiently small punctured disc it has no zeros. The corresponding closed coordinate-line disc lies in the chart and its boundary avoids B , hence lies in $\mathcal{G}$. Proposition 5.3 gives $x \in \mathcal{G}$, a contradiction. $\square$

The following consequence uses entire maps into the ambient manifold; their images need not avoid the analytic subset.

Corollary 5.5. *Let X be a complex manifold and $A \subset X$ a proper closed complex analytic subset. If X is dominable at every point of $X \setminus A$, then X is Oka.*

Proof. The case $\dim X = 0$ is immediate. For $n = \dim X \geq 1$, Lemma 2.3 gives

$$X \setminus A \subset \mathcal{G}, \quad X \setminus \mathcal{G} \subset A.$$

Since X is connected and A is proper, the germ of A is proper at every point. Corollary 5.4 yields $\mathcal{G} = X$, and Theorem 2.2 gives CAP. Thus X is Oka by Definition 1.1. $\square$

Proof of Theorem 1.5. The Oka manifold $X \setminus A$ is strongly dominable by Theorem 1.3. Its dominating entire maps, composed with the inclusion $X \setminus A \hookrightarrow X$, show that X is dominable at every point of $X \setminus A$. Corollary 5.5 applies. $\square$

6. AN APPLICATION TO SPACES OF RATIONAL FUNCTIONS

For $d \geq 1$, let

$$R_d = \text{Hol}_d(\mathbb{P}^1, \mathbb{P}^1)$$

be the manifold of rational maps of degree d . Hanysz [25] proved that R_2 is homogeneous and hence Oka, and that R_3 is strongly dominable and $\mathbb{C}$ -connected. The Oka property of R_3 was subsequently obtained in [33, Theorem 1.3]. For arbitrary degree, the Euclidean algorithm produces an Oka open set in the normalized evaluation fiber. The union theorem gives the Oka property of the whole fiber, and the evaluation bundle then recovers R_d .

For $f \in R_d$ satisfying $f(\infty) = 0$, the coprime representation $f = p/q$ has a unique normalization in which q is monic of degree d and $\deg p < d$. Thus the fiber of evaluation at infinity over 0 is

$$\mathcal{R}_d = \{(q, p) : q \text{ monic of degree } d, \deg p < d, \text{Res}(q, p) \neq 0\},$$

a smooth principal open subset of $\mathbb{C}^{2d}$.

Proposition 6.1. *For every $d \geq 1$, both $\mathcal{R}_d$ and R_d are Oka.*

Proof. The Euclidean algorithm supplies the coordinates of the Oka open set: linear partial quotients contribute parameters in $\mathbb{C}$, and normalization factors contribute parameters in $\mathbb{C}^*$. This is the case of linear partial quotients in Cazanave's continued-fraction decomposition [8, Lemma A.1]; its convention is obtained from ours by taking $(A, B) = (q, p)$ and considering q/p .

For $b_1, \dots, b_d \in \mathbb{C}$ and $a, c_1, \dots, c_{d-1} \in \mathbb{C}^*$, set

$$(6.1) \quad r_d = 1, \quad r_{d-1} = z - b_d, \quad r_{i-1} = (z - b_i)r_i + c_i r_{i+1} \quad (i = d-1, \dots, 1).$$

Then r_i is monic of degree $d - i$ and

$$\gcd(r_{i-1}, r_i) = \gcd(r_i, r_{i+1}) = 1,$$

by induction from $r_d = 1$. Consequently $(q, p) = (r_0, ar_1)$ belongs to $\mathcal{R}_d$.

Let $U_d \subset \mathcal{R}_d$ consist of the pairs for which $\deg p = d - 1$ and the successive nonzero monic remainders have degrees $d - 2, \dots, 0$. Monic division recovers the parameters in (6.1) uniquely:

$$a = \text{lc}(p), \quad r_0 = q, \quad r_1 = p/a, \quad r_{i-1} = (z - b_i)r_i + \tilde{r}_{i+1}, \quad c_i = \text{lc}(\tilde{r}_{i+1}), \quad r_{i+1} = \tilde{r}_{i+1}/c_i.$$

Here lc denotes the leading coefficient. At each division the coefficients are regular functions on the open set obtained at the preceding step, and the next degree condition is the nonvanishing of the coefficient of z^{d-i-1} in $\tilde{r}_{i+1}$. Hence U_d is Zariski open, and (6.1) and monic division give algebraic inverses establishing

$$(6.2) \quad U_d \cong \mathbb{C}^d \times (\mathbb{C}^*)^d.$$

In particular, U_d is nonempty and Oka, since complex Lie groups are Oka [14, Proposition 5.5.1]. The principal open set $\mathcal{R}_d \subset \mathbb{C}^{2d}$ is irreducible and connected. Thus (6.2) exhibits a dense Zariski open Oka subset, and Theorem 1.5 gives that $\mathcal{R}_d$ is Oka.

Finally, evaluation at infinity gives a holomorphic fiber bundle

$$(6.3) \quad \text{ev}_\infty : R_d \longrightarrow \mathbb{P}^1$$

with fiber $\mathcal{R}_d$. Indeed, near any target value a , choose Möbius transformations g_a depending holomorphically on a and satisfying $g_a(0) = a$. A local trivialization is

$$f \longmapsto (f(\infty), g_{f(\infty)}^{-1} \circ f).$$

Since $\mathbb{P}^1$ is a complex homogeneous manifold and hence Oka [14, Proposition 5.5.1], and $\mathcal{R}_d$ is Oka by the preceding argument, the Oka fiber bundle theorem [14, Theorem 5.5.4], with the correction in [15, errata to Theorem 5.6.5], applied to (6.3), gives that R_d is Oka. $\square$

STATEMENT ON THE USE OF ARTIFICIAL INTELLIGENCE

This manuscript developed from a substantial body of earlier drafts and working notes, discussions within our research group, and sustained interaction with generative artificial intelligence during the exploration, organization, and revision of the arguments. The authors reviewed the resulting text and take full responsibility for its mathematical content and final form.

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